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# 1 Introduction

A large literature argues that trust matters for aggregate economic outcomes, including long-run growth Algan and Cahuc 2010. At the micro level, Butler, Giuliano, and Guiso 2016 show that economic performance is not simply increasing in trust: income is maximized at intermediate levels. This paper starts from that result and asks whether a similar nonlinearity appears for realized household investment returns.

Although the link between trust and economic growth does not need much justificiation, the mechanism is inherent in finance environments since contracting between borrowers and lenders requires a delayed payoff through the *promise of future repayment*. If trust is too low, households may avoid high-return opportunities. This is not a surprising result given the documented equity premium puzzle. Moreover, if trust is too high, investors may under-monitor risk and prices and suffer otherwise avoidable. Frankly, if this mechanism results in a hump-shaped relationship between income and trust, then the pattern should be especially true in a setting which inherenntly requires trust between contracting parties such as returns.

Recent evidence using Norwegian administrative data documents persistent idiosyncratic return differences across households and scale dependence with wealth Fagereng et al. 2020. Related work using U.S. panel data also emphasizes a persistent return component and its relevance for inequality and life-cycle outcomes Daminato and Pistaferri 2024. These findings motivate asking what household characteristics can help explain the persistent component.

Replicating the Norwegian measurement environment in the U.S. is difficult. U.S. researchers typically combine survey data with partial administrative information, rather than observing a full administrative panel of wealth flows at the population level Kennickell 2017. I therefore use the HRS RAND Longitundinal 2022 version of the data, which is unusually useful for this question because it contains the core ingredients needed to construct returns over time: capital income, asset values across waves, debt, and transaction-related information as well as a measure of trust with nonmissing responses from a significant subset of the sample.

I verify whether the trust-income hump shape appears in the HRS and then test for a comparable hump shape regarding trust and returns to net wealth. Since trust is observed in one wave, I follow the literature and treat trust as time-invariant in pooled panel specifications Lesmeister, Limbach, and Goergen 2019.

The main results are as follows. First, I recover the hump-shaped relationship between trust and total income in the HRS. Second, for returns to net wealth, cross-sectional 2022 results are weaker, but the hump-shaped pattern appears in average-return and pooled panel specifications from 2002-2022, including after adding controls for risk exposure. Third, when moving to the fixed-effects decomposition and second-stage regression on time-invariant covariates, trust loses significance, while race and education remain predictive of the persistent component.

In extensions, I include financial literacy and alternative trust aggregates tied more directly to financial institutions. Financial literacy (especially the inflation question and literacy score) explains variation in returns in several specifications. For trust in financial institutions, evidence is mixed in cross-section and average specifications, but more favorable in panel settings, with some specifications again suggesting nonlinear trust effects.

The rest of the paper proceeds as follows. Section **??** reviews the literature on trust and returns. Section **??** describes the HRS data and construction of income, wealth, portfolio shares, and return measures. Section **??** presents the main empirical results for trust, income, and returns, including panel and fixed-effects specifications. Section **??** reports extension results with financial literacy

and trust in financial institutions.

# 2 Literature Review

## 2.1 Trust

There are three strands of the trust literature of interest to this paper. First, this paper draws from the literature regarding the detereminants of trust. An example of this is Alesina and Ferrara 2000. Key takeaway is that the standard demographic control variables (age, race, education, gender) as well as living in a community with much heterogeneity in terms of either race or income.

This paper also seeks to contribute to the trust literature regarding the empirical relationship between measures of trust and economic performance. A notable example relevant for this paper is Guiso, Sapienza, and Zingales 2008 which finds that trust can lead to greater stock market participation.

A related work Guiso, Sapienza, and Zingales 2004 finds that measures of social capital and trust leads to greater use of financial instruments like checks, as well as poortfolio diversification away from holding cash and towards holding stocks and other assets.

A paper which serves as a key motivation for this work is Butler, Giuliano, and Guiso 2016. They find that the relationship between trust and income is humped shaped; an intermediate level of trust is associated with the maximal level of income.

Haran Rosen, Lusardi, and Mitchell 2025 uses the specially-designed modules on trust from the HRS to study retirement security. They identify aggregate measures of trust; in particular, *Trust in Financial Institutions* is a measure of trust that should be relevant for returns, if they hump-shaped relationship with returns holds for the general trust measure. Additionally, the authors identify the set of financial literacy questions in the same wave of the survey, which I also make use of here.

Causal effects in the trust literature also inform this paper. Alsan and Wanamaker 2018 find that the 1972 public revelation of the Tuskegee Syphilis Study lead to lower trust in black males in medical institutions, ultimately leading to worse health outcomes over a lifetime.

Algan and Cahuc 2010 find that aggregate trust has a large positive causal effect on GDP per capita by showing that a measure of inherited trust by second generation immigrants in the U.S. is strongly correlated with aggregate trust in the origin country (and is persistent across generations).

## 2.2 Returns

An important work in the literature on empirical estimates of the rate of return is Fagereng et al. 2020 which characterizes the persistent component of returns for indivdiuals in Nowegian population data. This work also provides estimates of the distribution of returns within narrowly defined asset class, as well as for net wealth, which can serve as a useful benchmark for comparison.

Daminato and Pistaferri 2024 also documents the persistent component of returns in the PSID, and then use the data to calibrate an return-earnings process within a life cycle model of consumption-saving behavior.

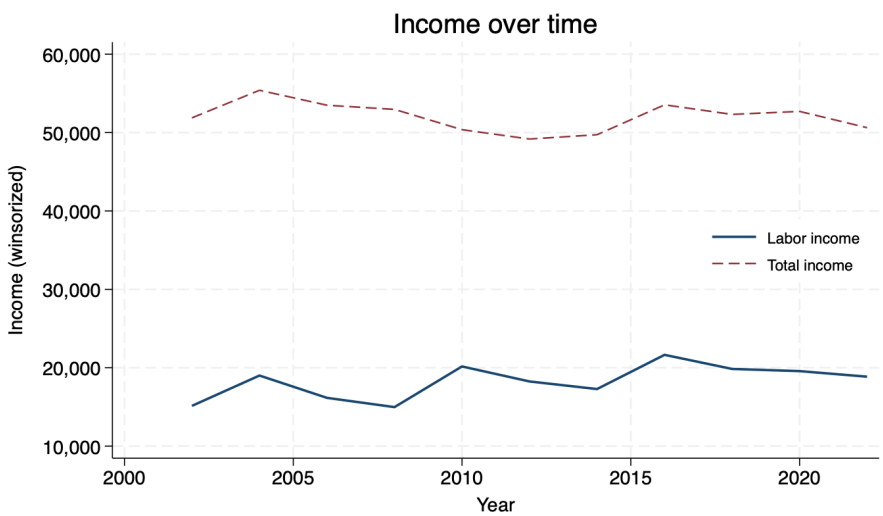
# 3 Descriptive statistics

For the statistical analysis in this paper, I use the RAND Longitudinal version of the Household Retirement Survey (HRS) public data. I focus on the years 2002-2022; as this time horizon was filled

with many economically relevant events (GFC, Covid), we can be creative regarding how to assess the accuracy of the data measured here. This is useful because, although there are many sources of income data to compare resuults to, there are less counterparts like this for return measures. The variables of interest fall into the categories: i) income, ii) wealth and portfolio composition, iii) returns, iv) trust, v) demographics, and vi) other controls.

## 3.1 Income

First, I use two measures of income: labor and total. Labor income is a narrow measure only capturing earnings and unemployment income. Total income is a more broad measure including retirement and capital income. Table **??** shows that, for the time period, mean incomes are relatively flat over the period. Moreover, the discrepancy between mean labor and total income is likely driven by the oversampling if older and retired households by the HRS.



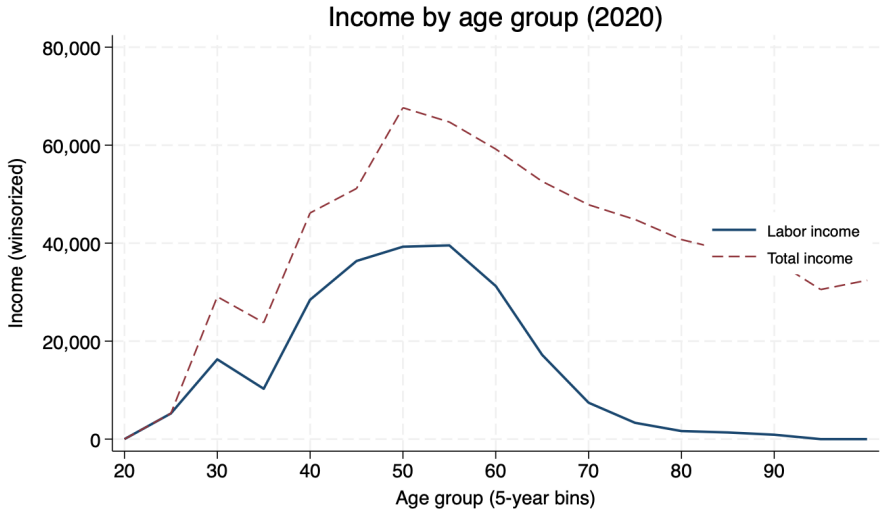

In particular, I pooled the observations of the survey together to get a sense of the distribution of incomes measured in the survey. This is captured by Table 3.1.

**Table 1:** Income distribution summary statistics (pooled waves)

| Variable | Obs | Mean | SD | P50 | P95 | Min | Max |
|---|---|---|---|---|---|---|---|
| Labor income (real, winsorized) | 204,952 | 18,330 | 37,032 | 0 | 97,924 | 0 | 215,261 |
| Total income (real, winsorized) | 204,952 | 52,004 | 68,036 | 29,103 | 177,498 | 0 | 457,548 |

Real USD, winsorized; summary over person-years.

An important finding in the literature on earnings in the U.S. is that it is generally hump-shaped over the lifecycle. It is a good sign then that figure **??**




Another empirical finding in the literature on earnings is that on average individuals with more education earn higher income. The trend captured in Table 3.1 suggests that the earnings data is in line with one would respect regarding earnings for a representative survey in the U.S.[1]

**Table 2:** Mean income by education group (real, winsorized)

| Education | Labor income (mean $) | Total income (mean $) | Obs |
|---|---|---|---|
| No hs | 6,075 | 23,987 | 44,654 |
| Hs | 12,590 | 39,601 | 63,826 |
| Some college | 20,298 | 53,483 | 47,390 |
| 4yr degree | 32,779 | 82,578 | 24,820 |
| Grad | 37,445 | 104,066 | 23,091 |

Real USD, winsorized. No hs = <12y; Hs = 12y; Some college = 13–15y; 4yr = 16y; Grad = 17+y.

[1] Although the HRS oversamples older households, I use the provided the respondent-level weights for this interpretation of the summary statistics of the data.

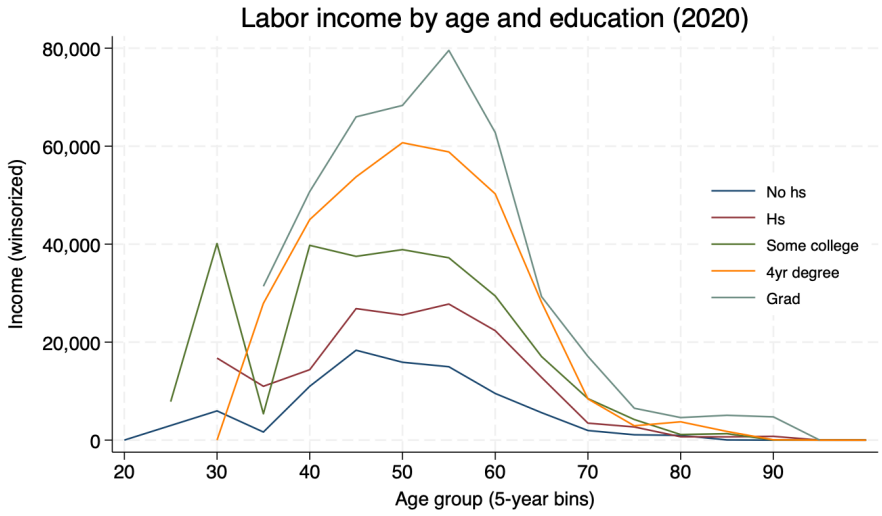



**Figure 1:** Labor income by age and education (2020)

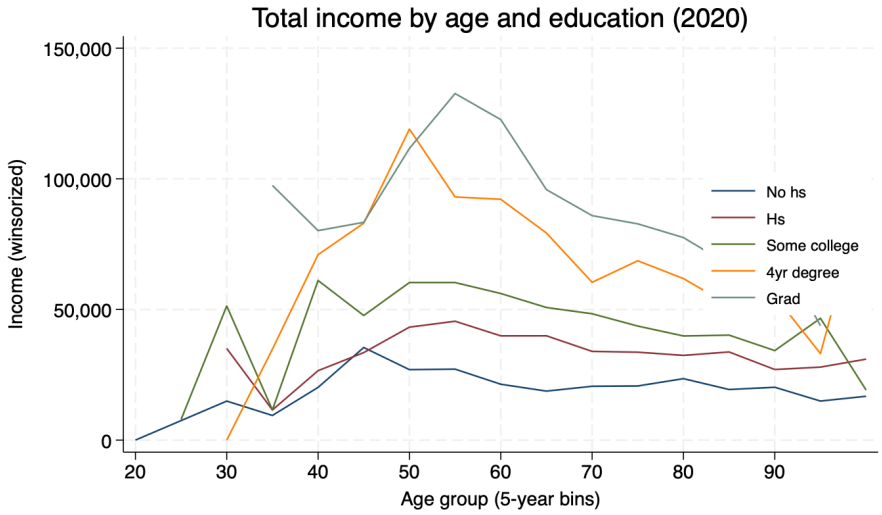



**Figure 2:** Total income by age and education (2020)

## 3.2 Asset classes and portfolio definitions

The HRS asks a number of questions aimed at measuring household wealth and portfolio composition in the sample. I collect data on i) **interest income and dividends**, ii) **capital gains**, iii) **net investment flows**, and iv) **previous period wealth holdings**, all of which are necessary

to define the desired measure of household returns. Since population-level administration data on these objects for individual taxpayers is not available in the U.S., it is important to understand each component needed in the return calculation. This will help to determine whether or not our measured distribution of returns in this dataset is sensible. To understand what portfolios are accruing returns, we will let the structure of the HRS questions guide how we define returns to specific asset classes (and collections of them) that the survey collects data for.

#### 3.2.1 Interest income and dividends

The HRS RAND longitudinal file has a distinctive way in which interest income and dividends received on major asset classes, like stocks and bonds, private business, and real estate, is measured. In particular, there is a variable in the survey which measures "household capital income":

- ...is the sum of household capital income received over the last calendar year, including business or farm income, self-employment earnings, gross rent, dividend and interest income, trust funds or royalties, and other asset income.

This particular feature of the data is the driving factor for the key modeling assumptions of this paper regarding measuring returns: asset classes are defined as narrowly as they can be given observation of interest income and dividends on those assets. For example, although we see capital gains for stocks, business, bonds, and real estate, we do not observe interest income or dividends on these assets individually. Thus, the narrowest asset class defined in this paper will be called "core assets" and will be comprised of these assets.

The RAND version of the data offers a number of variables measuring pension and annuity income as well as other forms of retirement income. I use this to construct a measure of returns to retirement assets and to define a broader notion of portfolio returns by considering returns to core and retirement assets.

A key assumption is that there are no interest income or dividends earned on residential assets[2]. With this assumption, we can consider interest income and dividends on the entire portfolio as the same as interest income and dividends on the portfolio with just core and retirement assets. Thus, to compute a measure of returns to net wealth, I need to add in the remaining available capital gains per asset class (for those that receive no interest income).

[2]Or on any of the asset classes for which capital gains are available for.

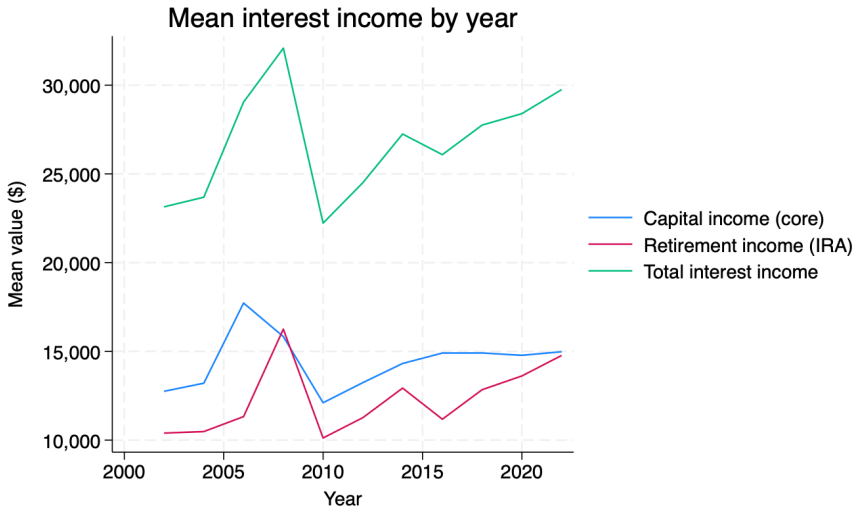



**Figure 3:** Interest income mean by year

### 3.2.2 Capital gains

The possible asset classes for which capital gains can be computed in the HRS are i) primary residences, ii) secondary residences, iii) other real estate, iv) private business, v) IRA/Keogh (or "retirement"), vi) stocks/mutual funds, vii) bonds, viii) checking/savings/money market, ix) cds/t-bills, x) vehicles, xi) other assets. The possible liabilities are i) mortgages on primary residence, ii) mortgage on secondary residence, iii) other home loans, and iv) total other debt.

Capital gains for each asset class are computed as the estimated change in valuation across survey waves. The general trend is that these changes move substantially over time. In the panel (Figure 4), business, stocks, and other real estate all show large cyclical swings and end the sample at levels above where they begin, while safe assets (bonds + checking/savings + CDs/T-bills) are flatter and lower-volatility by comparison.

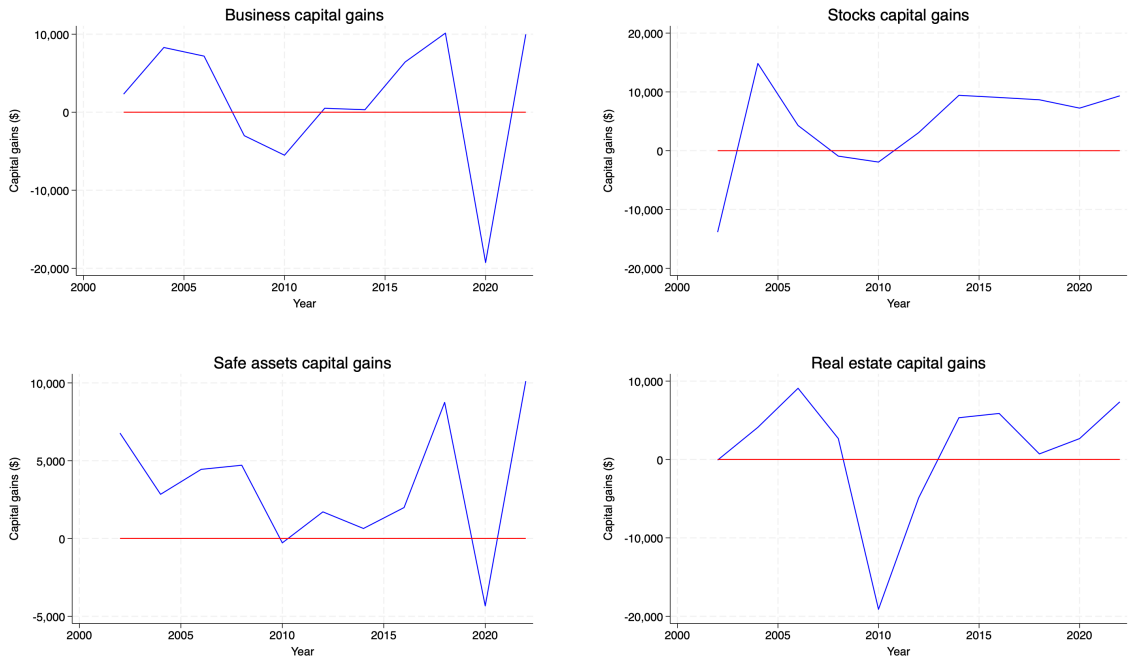



**Figure 4:** Capital gains in core assets, by year

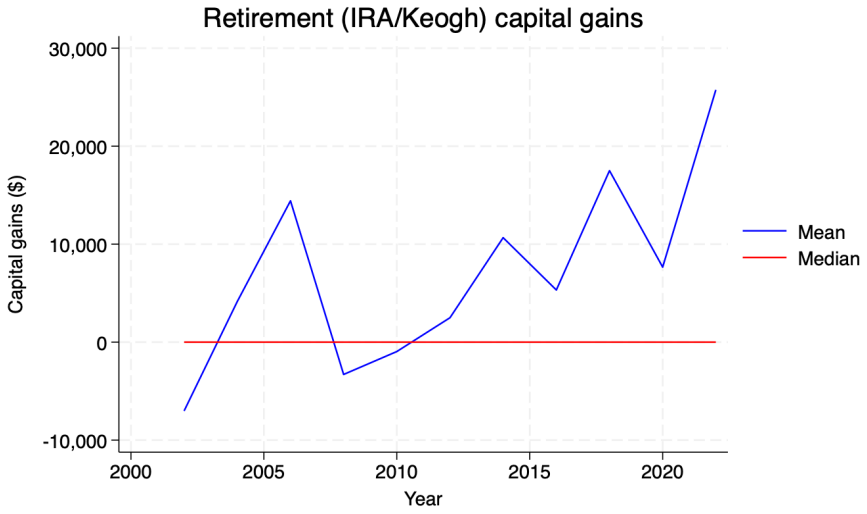



**Figure 5:** Capital gains in retirement assets, by year

Another feature that is apparent by taking a look at capital gains is that, most respondent hold

no assets in a particular class. This can be seen by the red median line at 0 for most waves. In the residential panel (Figure 6), the primary-residence series is much larger in level and volatility than the secondary-residence series, and only primary residence shows a clearly positive median in some years. This is another sign that the data is sensible so far – large amount of non-participation despite evidence of returns is consistent with the empirically documented equity premium puzzle.

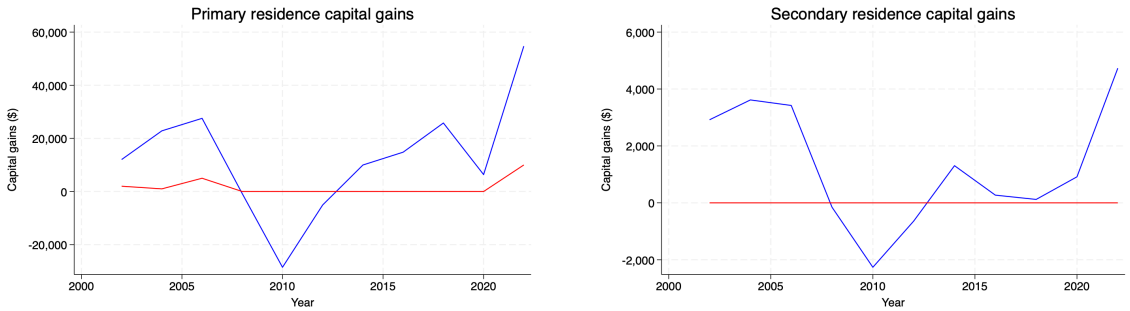



**Figure 6:** Capital gains in residential assets, by year

Notably, the synchronized dip around 2008–2010 for primary residence, secondary residence, and other real estate is also a good sign regarding how reliably measured the data is.

### 3.2.3 Net investment flows

Net investment flows are vital in computing returns accurately because capital gains and interest income can miss other flows of value into a given asset class. In the HRS dataset, net investment flows are the only variable relevant to the returns calculation that RAND did not clean and process. For this reason, there are substantially less observation for this variable per asset class than the others. To work around this, I do two things. First, I process the net investment flow per asset class myself by using the associated flag variable (asking yes or no if an individual has "bought or sold since the previous wave") for each variable. Second, I assume that if an individual receives interest income on that asset class, but their flow in nonmissing, then the nonmissing flow to that asset class is treated as 0.[3]

With this in mind, I present a figure of net investment flows into the assets available in the dataset. As you can see, the magnitudes for flows into a given class are comparable. I use these flows to construct net investment flows into i) **core assets**, ii) **retirement assets**, iii) **residential assets**, iv) **core and residential assets**, and v) **net wealth**.

[3]Non-missing interest income (and capital gains) indicate participation within an asset class.

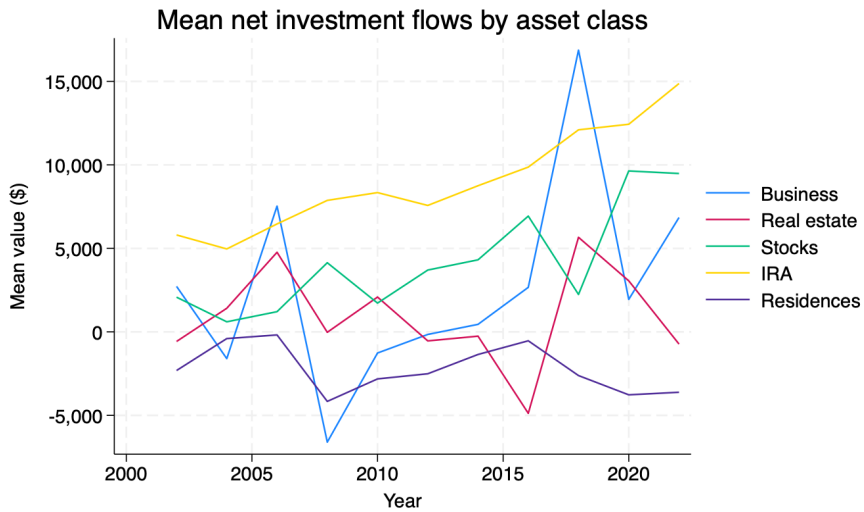



**Figure 7:** Flows by asset, mean by year

### 3.2.4 Wealth and inequality

After defining asset classes, I consider mean wealth within these narrowly defined asset classes for each year of the sample. Interestingly, retirement assets seem to perform the worst in terms of average returns. Core assets seem to offer the best performance, and all asset classes see a downward dive leading up to 2010 – historically accurate in the context of investment performance during the global financial crisis.

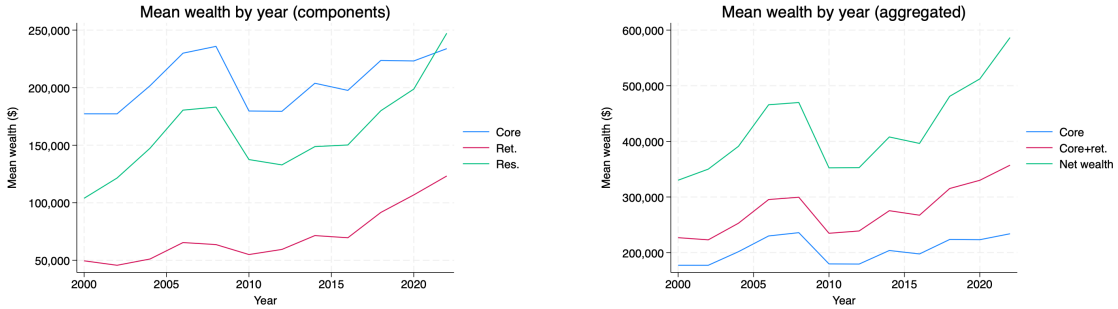



**Figure 8:** Mean wealth by year

I turn attention to measures of wealth in the sample based on the portfolio definitions: i) net wealth in core assets, ii) net wealth in core and retirement assets, iii) total net wealth.

I consider the distribution of these measures of income and wealth across respondents of the survey in the following figures. For income, the measure of labor income is significantly more unequally distributed than the measure of total income in the sample. This is likely an artifact

of the sampling method of the HRS (older likely overrepresented and are towards the end of their working years).

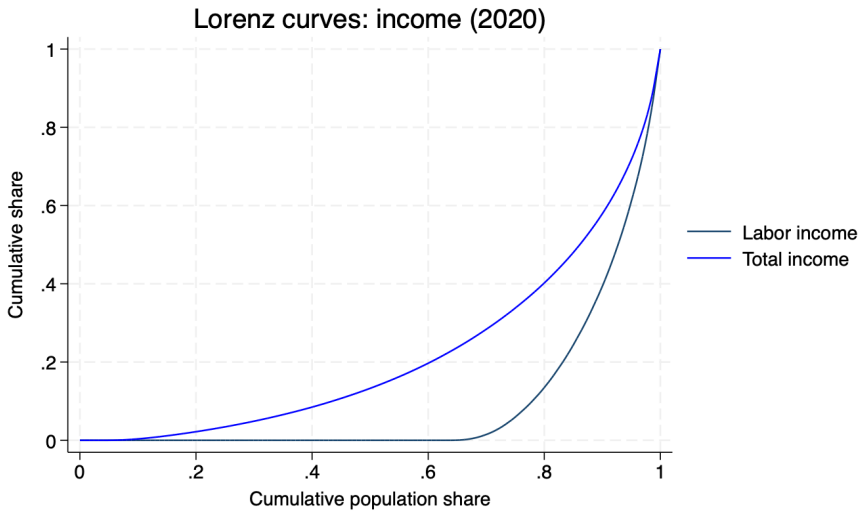



**Figure 9:** Lorenz: income (2020)

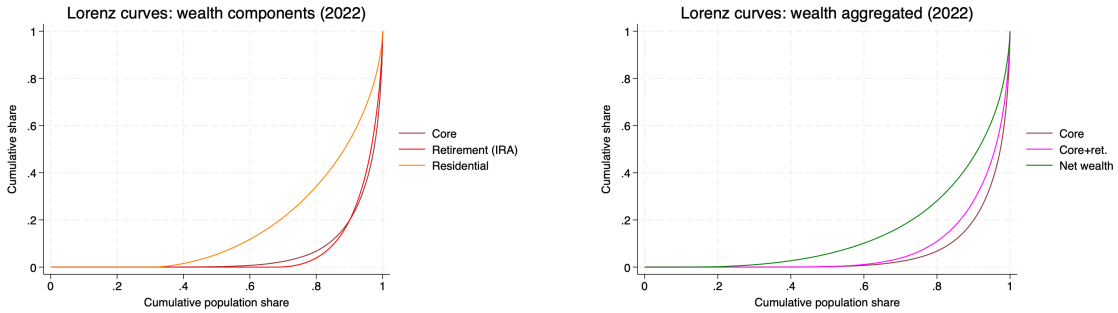



**Figure 10:** Lorenz: wealth (2022)

For wealth, holdings in core assets and retirement assets are more unequally distributed than holdings in residential assets. This is in line with common knowledge that home ownership is the most common form of asset ownership in the U.S.

Interestingly, the distribution of net wealth become less unequal when it is extended to incorporate retirement assets on top of core assets. this can be seen in the following figure .

### 3.2.5 Portfolio composition

To describe the composition of portfolio during the period, I document mean portfolio shares for the relevant asset classes.

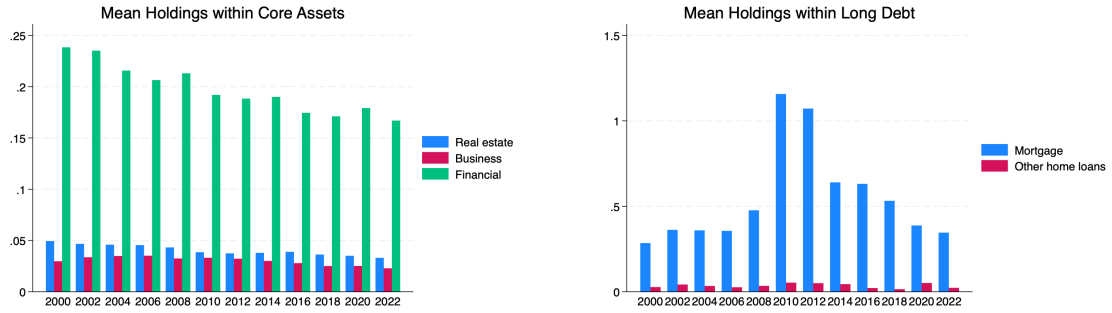



**Figure 11:** Mean holdings within assets and liabilities, by year

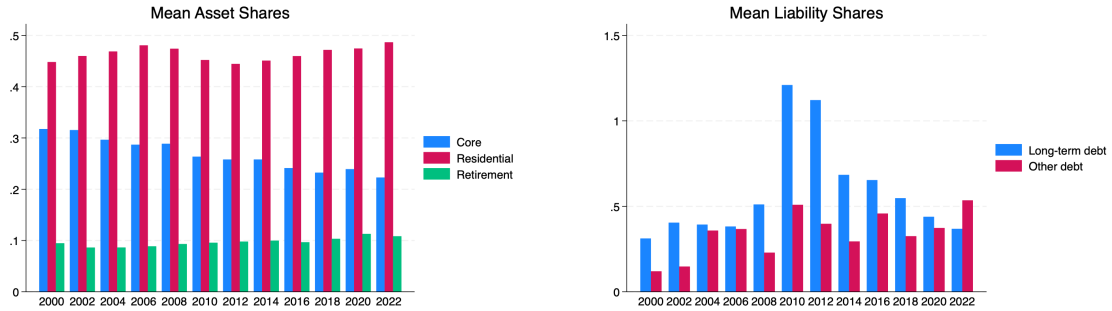



**Figure 12:** Mean asset and debt shares, by year

Investment behavior is likely to vary with income and wealth. Asset shares (core, residential, retirement) by percentile of labor income, total income, and net wealth (2020):

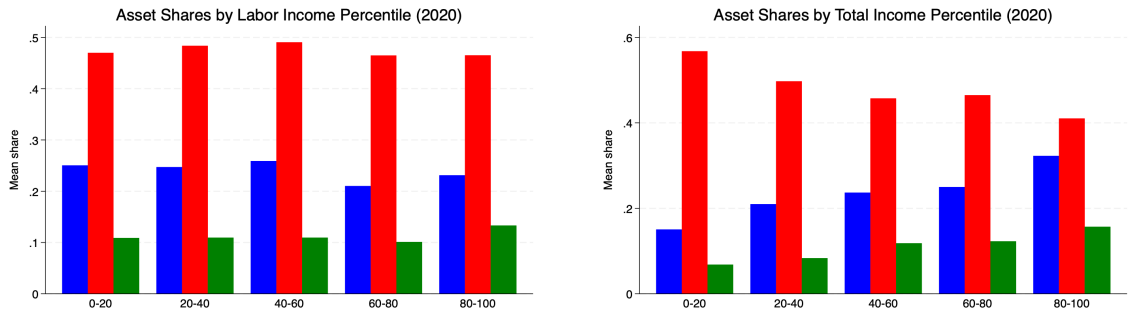



**Figure 13:** Asset shares by labor and total income percentiles (2020)

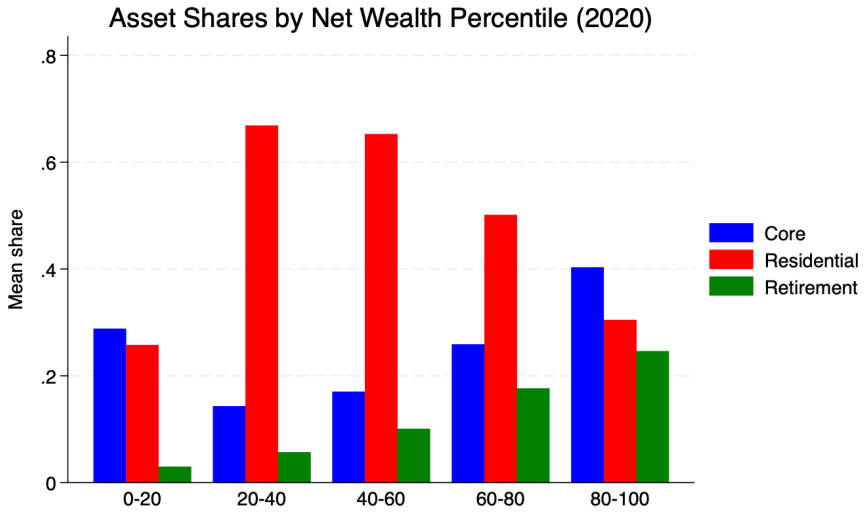



**Figure 14:** Asset shares by net wealth percentile (2020)

We can also look at the same breakdown by income and wealth percentiles but for the ratio of debt to gross assets. This is captured in the following figures .

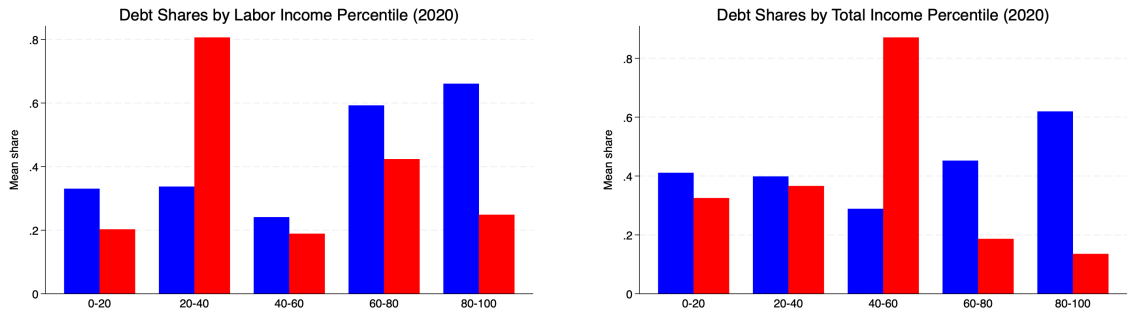



**Figure 15:** Debt shares by labor and total income percentiles (2020)

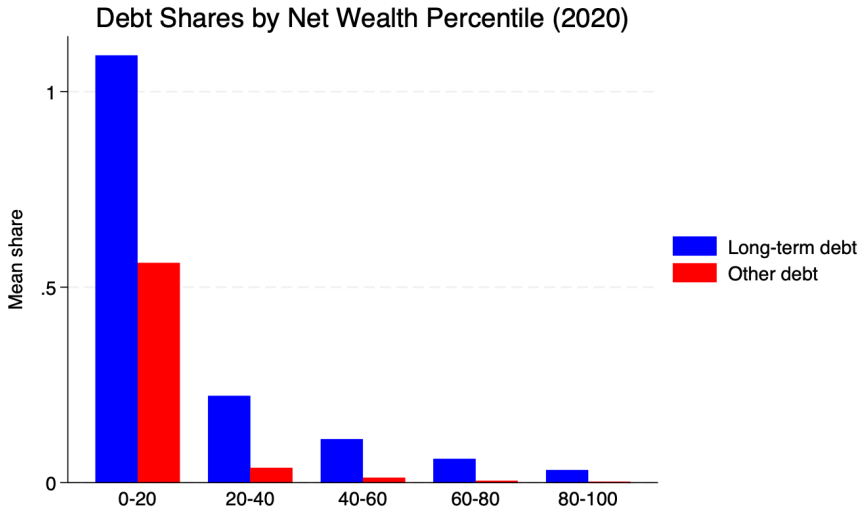



**Figure 16:** Debt shares by net wealth percentile (2020)

I summarize pooled participation and share statistics in Table 3.2.5.

**Table 3:** Participation and portfolio composition

| Statistic | Mean | SD | P1 | P50 | P99 | Obs |
|---|---|---|---|---|---|---|
| Fraction with business assets | 0.093 | 0.290 | 0.000 | 0.000 | 1.000 | 205,771 |
| Fraction with stocks assets | 0.262 | 0.440 | 0.000 | 0.000 | 1.000 | 205,771 |
| Fraction with safe assets | 0.827 | 0.378 | 0.000 | 1.000 | 1.000 | 205,743 |
| Fraction with other real estate assets | 0.149 | 0.356 | 0.000 | 0.000 | 1.000 | 205,771 |
| Fraction with retirement assets | 0.384 | 0.486 | 0.000 | 0.000 | 1.000 | 205,769 |
| Fraction with residential assets | 0.808 | 0.394 | 0.000 | 1.000 | 1.000 | 205,769 |
| Fraction with core assets | 0.859 | 0.348 | 0.000 | 1.000 | 1.000 | 205,743 |
| Fraction with long-term debt | 0.363 | 0.481 | 0.000 | 0.000 | 1.000 | 205,771 |
| Fraction with other debt | 0.335 | 0.472 | 0.000 | 0.000 | 1.000 | 205,766 |
| Fraction with some debt | 0.533 | 0.499 | 0.000 | 1.000 | 1.000 | 205,766 |
| Share in private business | 0.030 | 0.121 | 0.000 | 0.000 | 0.683 | 205,771 |
| Share in stocks/mutual funds | 0.059 | 0.151 | 0.000 | 0.000 | 0.741 | 205,771 |
| Share in safe assets | 0.116 | 0.210 | 0.000 | 0.032 | 1.000 | 205,743 |
| Share in other real estate | 0.041 | 0.131 | 0.000 | 0.000 | 0.683 | 205,771 |
| Share in retirement assets | 0.096 | 0.184 | 0.000 | 0.000 | 0.798 | 205,769 |
| Share in residential assets | 0.463 | 0.344 | 0.000 | 0.454 | 1.000 | 205,769 |
| Share in core assets | 0.270 | 0.301 | 0.000 | 0.143 | 1.000 | 205,743 |
| Share in core and retirement assets | 0.366 | 0.330 | 0.000 | 0.306 | 1.000 | 205,741 |
| Leverage, Long-term debt | 0.601 | 5.921 | 0.000 | 0.000 | 8.583 | 205,771 |
| Leverage, Other debt | 0.343 | 11.333 | 0.000 | 0.000 | 5.000 | 205,766 |
| Leverage, Total debt | 0.944 | 12.888 | 0.000 | 0.005 | 14.330 | 205,766 |

I also report pooled composition by wealth fractiles for the core distribution and for net wealth.

**Table 4:** Asset composition by wealth

| Fractile | Real est. | Business | Stocks | Safe assets | Mortgage debt | Other home loans | Other debt |
|---|---|---|---|---|---|---|---|
| Bottom 10% | 0.000 | 0.000 | 0.000 | 0.000 | 1.073 | 0.031 | 0.678 |
| 10–20% | 0.000 | 0.000 | 0.000 | 0.001 | 1.099 | 0.017 | 2.471 |
| 20–50% | 0.002 | 0.001 | 0.004 | 0.140 | 1.059 | 0.080 | 0.725 |
| 50–90% | 0.055 | 0.035 | 0.077 | 0.142 | 0.173 | 0.013 | 0.024 |
| 90–95% | 0.130 | 0.106 | 0.208 | 0.110 | 0.037 | 0.004 | 0.004 |
| 95–99% | 0.162 | 0.151 | 0.245 | 0.093 | 0.029 | 0.004 | 0.003 |
| 99–99.9% | 0.193 | 0.233 | 0.262 | 0.074 | 0.014 | 0.003 | 0.003 |
| 99.9–99.99% | 0.255 | 0.363 | 0.173 | 0.063 | 0.008 | 0.001 | 0.001 |
| Top 0.01% | 0.105 | 0.576 | 0.203 | 0.057 | 0.002 | 0.000 | 0.000 |

**Table 5:** Portfolio composition by wealth

| Fractile | Core | Residential | Retirement | Long debt | Other debt | Total debt |
|---|---|---|---|---|---|---|
| Bottom 10% | 0.225 | 0.352 | 0.033 | 2.938 | 1.460 | 4.398 |
| 10–20% | 0.408 | 0.128 | 0.011 | 1.196 | 1.070 | 2.267 |
| 20–50% | 0.186 | 0.606 | 0.050 | 0.262 | 0.060 | 0.322 |
| 50–90% | 0.301 | 0.483 | 0.148 | 0.064 | 0.007 | 0.071 |
| 90–95% | 0.460 | 0.290 | 0.202 | 0.028 | 0.002 | 0.030 |
| 95–99% | 0.535 | 0.235 | 0.184 | 0.022 | 0.002 | 0.024 |
| 99–99.9% | 0.646 | 0.190 | 0.124 | 0.015 | 0.002 | 0.017 |
| 99.9–99.99% | 0.689 | 0.200 | 0.068 | 0.007 | 0.000 | 0.007 |
| Top 0.01% | 0.477 | 0.126 | 0.217 | 0.003 | 0.000 | 0.003 |

## 3.3 Returns

I follow recent literature using Norwegian administrative tax data at the population level and and using PSID in terms of the components used to construct a measure of returns. However, I take seriously the structure of the dataset I am working with(the HRS RAND longitudinal file) in defining the portfolio/asset class for which the returns are being accrued to. As mentioned before, core assets is the most narrowly defined asset class here because interest income and dividends does not disaggregate into the asset classes: stocks, bonds, private business, IRA/Keogh.

The following formula from Daminato and Pistaferri 2024 serves as motivation for the return measures defined using the HRS data:

$$r_t = \frac{y_t^c + cg_t - F_t - y_t^d}{A_{t-1} + .5F_t}$$

where $y_t^c$ interest income and dividends, capital gains $cg_t$ measured as the difference between reported stock across waves, $F_t$ net investment flows, $y_t^d$ payments on debt (in the RAND longitudinal file, the variables were mentioned are all in net terms so this variable was 0), and $A_{t-1}$ total net wealth at beginning of previous period.

### 3.3.1 Narrow asset classes

Recall that we only see interest income and dividends for core assets and retirement assets. Since this is an important component of the returns calculation, we construct the narrowest return measure on core assets. To do so, I keep track of all asset classes captured in the capital gains variable[4] and aggregate the capital gains and net investment flows for this group of assets.

Returns to core assets are given by

$$r_{i,t}^{core} = \frac{y_{i,t}^{core} + cg_{i,t}^{core} - F_{i,t}^{core}}{A_{i,t-1}^{core} + .5F_{i,t}^{core}}$$

where $y_{i,t}^{core}$ is the interest income and dividents on the core assets, $cg_{i,t}^{core}$ are the capital gains on the core assets[5], $F_{i,t}^{core}$ are the net investment flows on core assets, and importantly, $A_{i,t-1}^{core}$ is the total value of the core assets in the previous period.

The rest of the return measures are defined similarly. In the HRS data, we can see income on retirement assets as well. This allows us to define a return measure for retirement assets:

$$r_{i,t}^{ret} = \frac{y_{i,t}^{ret} + cg_{i,t}^{ret} - F_{i,t}^{ret}}{A_{i,t-1}^{ret} + .5F_{i,t}^{ret}}$$

where $y_{i,t}^{ret}$ is the interest income and dividents on the retirement assets, $cg_{i,t}^{ret}$ are the capital gains on the retirement assets, $F_{i,t}^{ret}$ are the net investment flows on retirement assets, and $A_{i,t-1}^{ret}$ is the total value of the retirement assets in the previous period.

Based on the modules offered in a given wave of the survey, assume that there are no interest income or dividends to either primary and secondary residential assets, so that the interest income

[4]Those asset classes are: other real estate, business/farm, stocks/mutual funds, bonds, checking/savings, and CDs/t-bills.

[5]where the capital gains for each asset class is $V_{i,t} - V_{i,t-1}$, where $V_{i,t}$ is the value of the asset (business or stock or real estate) for the i-th respondent in the t-th wave of the survey.

and dividends accrued to residential assets is 0 in a given period ($y_{i,t}^{res} = 0$). With this in mind, returns to residential assets are defined as:

$$r_{i,t}^{res} = \frac{cg_{i,t}^{res} - F_{i,t}^{res}}{A_{i,t-1}^{res} + .5F_{i,t}^{res}}$$

where $cg_{i,t}^{res}$ are the capital gains on the residential assets, $F_{i,t}^{res}$ are the net investment flows on residential assets (including improvement costs), and $A_{i,t-1}^{res}$ is the total value of the residential assets in the previous period.

### 3.3.2 Broad asset classes

After defining returns for narrow asset classes (i.e. returns to a single asset), I also define return measure for portfolios (returns on multiple assets). The first is for returns to both core and retirement assets:

$$r_{i,t}^{coret} = \frac{y_{i,t}^{coret} + cg_{i,t}^{coret} - F_{i,t}^{coret}}{A_{i,t-1}^{coret} + .5F_{i,t}^{coret}}$$

where $y_{i,t}^{coret}$ is the interest income and dividends on core and retirement assets, $cg_{i,t}^{coret}$ are the capital gains on core and retirement assets, $F_{i,t}^{coret}$ are the net investment flows on core and retirement assets, and $A_{i,t-1}^{coret}$ is the total value of core and retirement assets in the previous period.

Returns to net wealth is given by

$$r_{i,t}^{net} = \frac{y_{i,t}^{net} + cg_{i,t}^{net} - F_{i,t}^{net}}{A_{i,t-1}^{net} + .5F_{i,t}^{net}}$$

where $y_{i,t}^{net}$ is the interest income and dividends on core and retirement assets, $cg_{i,t}^{net}$ are the capital gains on all available asset classes in the survey, $F_{i,t}^{net}$ are the net investment flows on assets for which it is relevant in the survey (stocks, bonds, business, IRA, residences, and real estate), and $A_{i,t-1}^{net}$ is the total value of the available assets minus liabilities in the previous period[6].

I present the means for the return measures in the following figure 17. I group them by the returns at the asset level (core, retirement, residential) and at the portfolio level (core, core and retirement, net wealth).

[6] Mortgages on secondary residences are left out of the list of liabilities since it was an optional question in some waves of the survey.

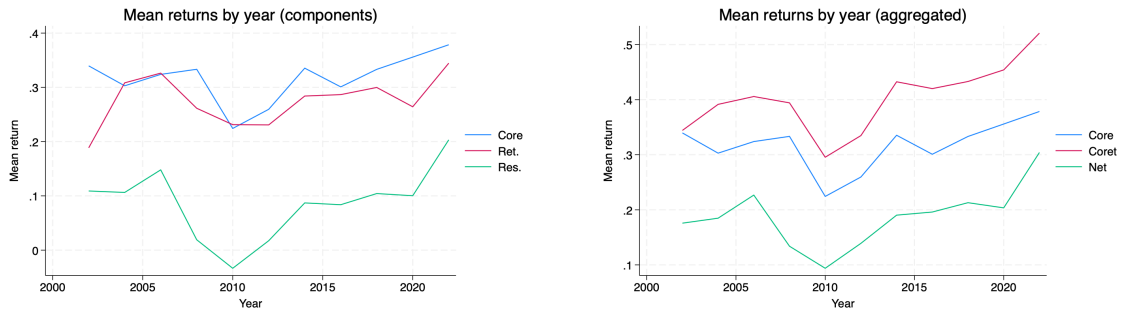



**Figure 17:** Mean returns by year

Figure 18 shows the distribution of the return measures across respondents by a similar grouping. In general, this attempt is encouraging, as its shape (in particular, for returns to net wealth) closely resembles early estimates of the empirical distribution of individual realized returns in the Norwegian population data.

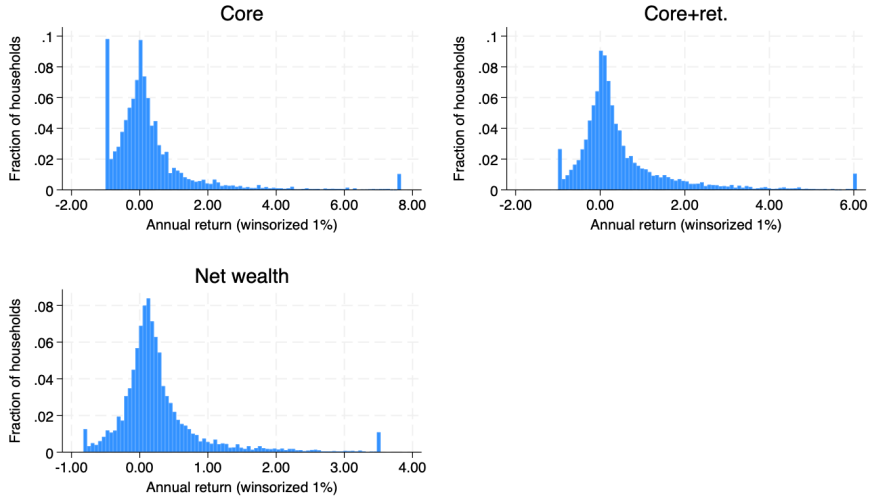



**Figure 18:** Returns histogram, aggregated (2022)

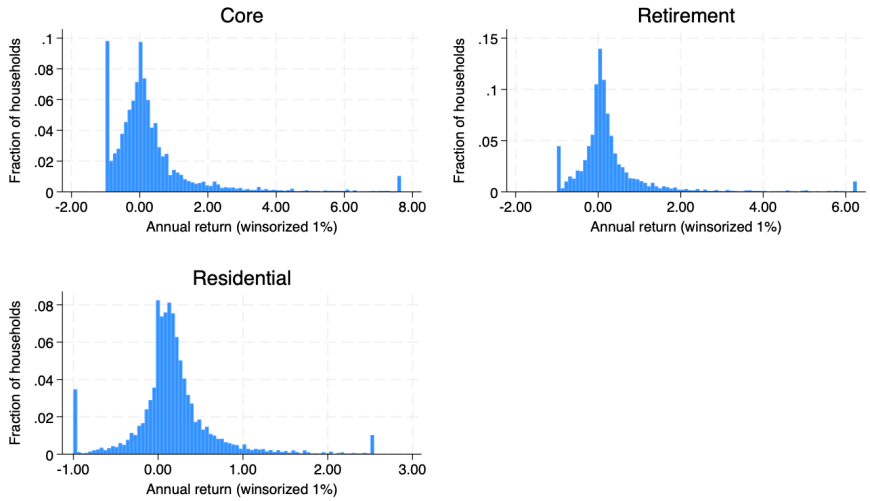



**Figure 19:** Returns histogram, components (2022)

**Table 6:** Returns to assets/portfolios

| Portfolio | Mean | SD | Skewness | Kurtosis | P1 | P50 | P99 |
|---|---|---|---|---|---|---|---|
| Core assets | 0.3144 | 1.6144 | 7.75 | 141.08 | -1.0000 | 0.0060 | 6.9373 |
| Residential assets | 0.0829 | 0.5502 | 5.87 | 101.59 | -1.0000 | 0.0351 | 2.0332 |
| Retirement assets | 0.2725 | 1.2087 | 7.26 | 141.05 | -1.0000 | 0.0550 | 5.1884 |
| Core and retirement assets | 0.3962 | 1.1969 | 6.43 | 143.91 | -1.0000 | 0.1087 | 5.0139 |
| Net wealth | 0.1841 | 0.6963 | 10.75 | 444.56 | -0.7831 | 0.0700 | 2.7839 |

## 3.4 Income and wealth

Income and wealth are understood to be positively correlated. To see this play out in the HRS data, I first look at mean labor income by wealth percentile. In ths figure, it is clear that a positive relationship seems to hold.

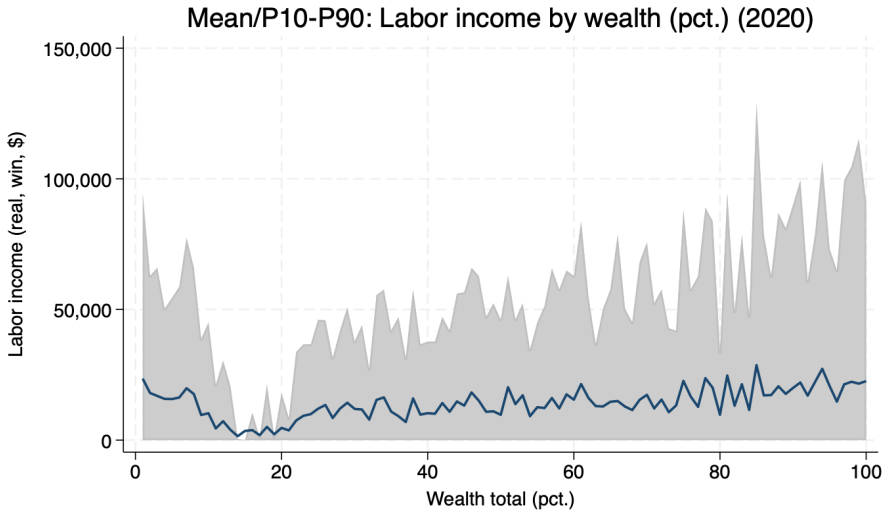



**Figure 20:** Labor income mean/P10-P90 by wealth percentile (2020)

The literature on income and wealth also documents significantly more inequality in the the upper tails of the wealth distribution than in the income distribution. The observations towards to top and bottom of the wealth distribution in the following figure seem to suggest a non-linear relationship between the variables. This pattern can be better seen in the figure 21.

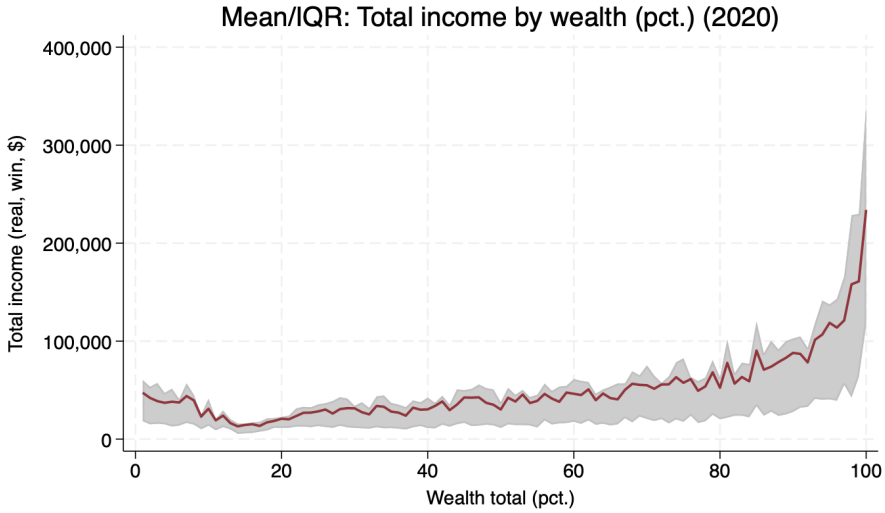



**Figure 21:** Total income mean/IQR by wealth percentile (2020)

## 3.5 Returns and wealth by portfolio

A positive relationship between wealth and returns has also been documented, referred to as *scale dependence*. In the next three figures, I show that there does seem to be a positive statistical relationship between the return measure (core, retirement, residential) and wealth.

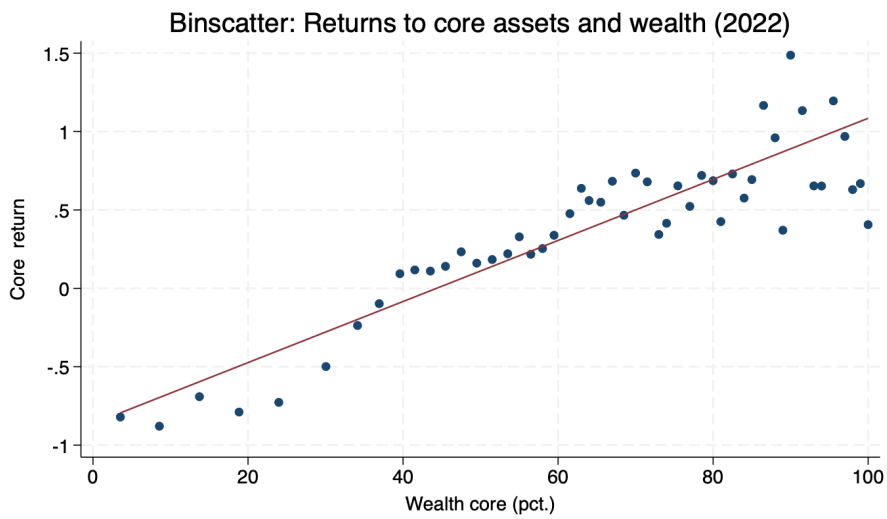



**Figure 22:** Core return binscatter (2022)

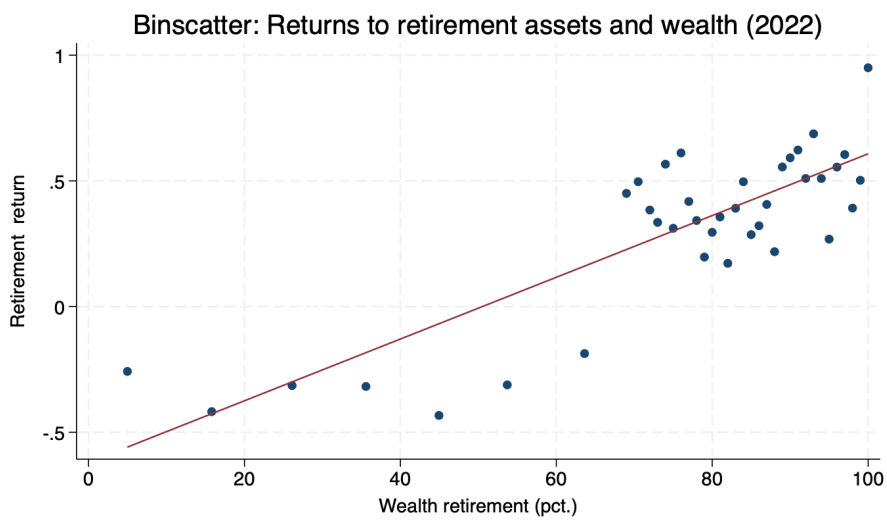



**Figure 23:** Retirement return binscatter (2022)

**Figure 24:** Residential return binscatter (2022)

I also show the positive statistical relationship between returns and wealth at the portfolio level, when comprised of core assets and retirement assets.

**Figure 25:** Core+IRA return binscatter (2022)

Interestingly, the relationship between wealth and returns to total net wealth does not appear positive. In fact, they correlation is slightly downward-sloping.

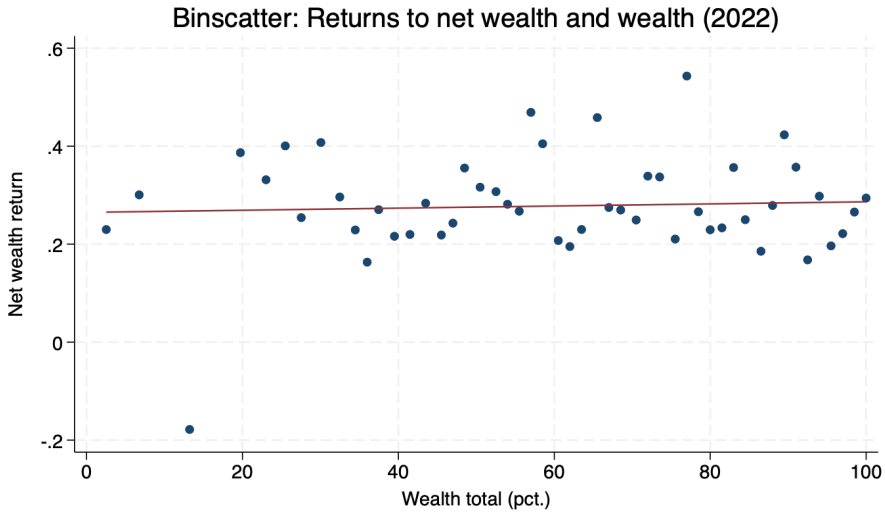



**Figure 26:** Net wealth return binscatter (2022)

## 3.6 Trust

I turn my attention to the suite of trust questions in "Section V: Modules" section of the 2020 HRS data. There are a total of 8 questions, asking respondents to say on a scale of 1-10 "how much do you trust people in general?" and of their trust in other features of American life relating to healthcare, finance, and media. These questions were only asked for a single year. Although this is an issue, because the panel structure of the HRS allows me to estimate the persistent component of returns, which is an important object in this literature. That said, in the pooled setting, we can assume that trust, like education, is fixed over time.[7]

I summarize general trust by age, gender, race, and education in Table 3.6.

[7] Literature on trust talks about how history of being cheated or treated fairly form individuals' trust over time. If this is true and at some point, one learns enough an forms their trust level, then a sample with an overrepresentation of older household is a reasonable environment to assume constant trust levels.

**Table 7:** General trust by demographics (2020)

**Panel A: By age**

| Group | N | Mean |
|---|---|---|
| 25-49 | 20 | 4.500 |
| 50-54 | 37 | 4.541 |
| 55-59 | 183 | 4.765 |
| 60-64 | 180 | 5.344 |
| 65-69 | 181 | 5.210 |
| 70-74 | 112 | 5.955 |
| 75-79 | 81 | 5.914 |
| 80-84 | 79 | 6.114 |
| 85-89 | 27 | 7.259 |

**Panel B: By gender**

| Group | N | Mean |
|---|---|---|
| Male | 388 | 5.446 |
| Female | 512 | 5.365 |

**Panel C: By race**

| Group | N | Mean |
|---|---|---|
| White (NH) | 446 | 5.951 |
| Black (NH) | 227 | 4.577 |
| Hispanic | 162 | 5.043 |
| Other (NH) | 64 | 5.359 |

**Panel D: By education**

| Group | N | Mean |
|---|---|---|
| No hs | 192 | 4.979 |
| Hs | 230 | 5.439 |
| Some college | 239 | 5.385 |
| 4yr degree | 121 | 5.760 |
| Grad | 114 | 5.667 |

I am interested in the relationship between trust and economic performance as the literature is, however in my setting the measure of economic performance is the return to assets. If a hump-shaped relationship is reasonable for income, it is even more plausible for returns: there is an inherent (and possibly explicit) level of trust between borrower and lender when forming credit contracts.

### 3.6.1 Trust and income

As discussed before, the literature on trust and economic performance suggests there's a "right amount of trust". I want to see if this hump shaped relationship holds between the measures of

trust and income in the HRS. After deflating and, more importantly, winsorizing labor income at the top and bottom 1%, the relationship seems to be hump shaped.

Income in the HRS is highly right-skewed and features a large mass at zero labor income (reflecting retirement and non-participation). In applied work, a common baseline is to use log(income) because it reduces the influence of extreme values and yields coefficients interpretable as semi-elasticities. However, $\log(\cdot)$ is undefined at zero (and problematic with negative values when they arise from measurement or netting conventions), so using log(income) in the HRS mechanically drops a large fraction of observations and can substantially change the visual and econometric relationship.

In our data, the corresponding scatter of log(income) versus trust changes markedly because so many observations have exactly zero labor income. Using $\log(1+\text{income})$ retains zeros but imposes an ad hoc curvature near zero and, in our case, does not restore the same qualitative picture.

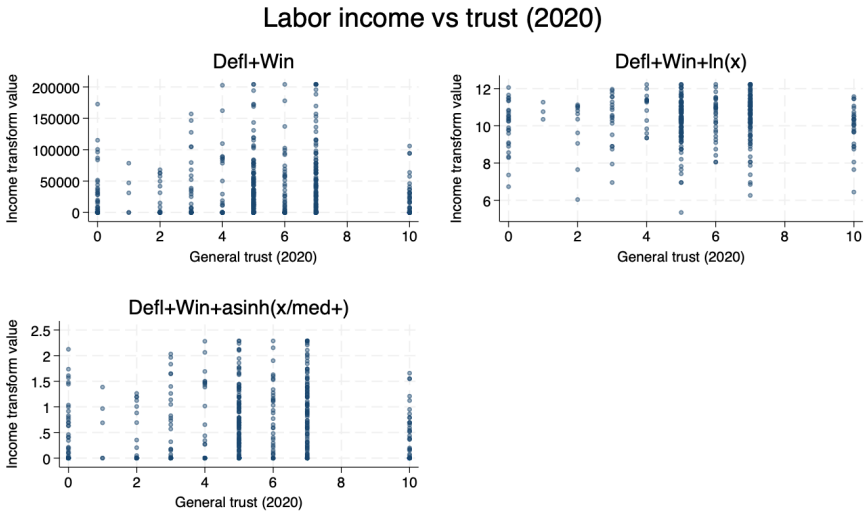



**Figure 27:** Labor income measures vs. trust (2020)

Motivated by this issue, we follow a large applied econometrics literature that uses the inverse hyperbolic sine (IHS, or asinh) transformation as a "log-like" mapping that is well-defined at zero and can accommodate negative values. The IHS transformation was advocated early on as an alternative to the log for heavy-tailed outcomes with extreme values and potential zeros/negatives (Burbidge, Magee, and Robb, *"Alternative Transformations to Handle Extreme Values of the Dependent Variable"*; MacKinnon and Magee, *"Transforming the Dependent Variable in Regression Models"*). More recently, Bellemare and Wichman (*"Elasticities and the Inverse Hyperbolic Sine Transformation"*) provide practical guidance on interpretation, emphasizing that $\text{asinh}(y)$ behaves similarly to $\log(y)$ for large $y$ (since $\text{asinh}(y) \approx \log(2y)$ when $y$ is large) while remaining defined at $y = 0$.

A related point is that the IHS can be sensitive to the units of measurement, which motivates a *scaled* IHS of the form $\text{asinh}(y/\kappa)$ (see discussions in Aïhounton and Henningsen, *"Units of Measurement and the Inverse Hyperbolic Sine Transformation"*; Norton, *"The Inverse Hyperbolic Sine*

*Transformation*"). In practice, $\kappa$ is chosen as a meaningful scale (e.g., the median positive income), which makes the transformation more comparable across samples and improves interpretability near zero. Empirically, applying the (scaled) IHS to our deflated and winsorized income restores a scatter with trust that is visually well-behaved while retaining the economically important mass at zero labor income. This can be seen for the measure of toal income as well in the following figure .

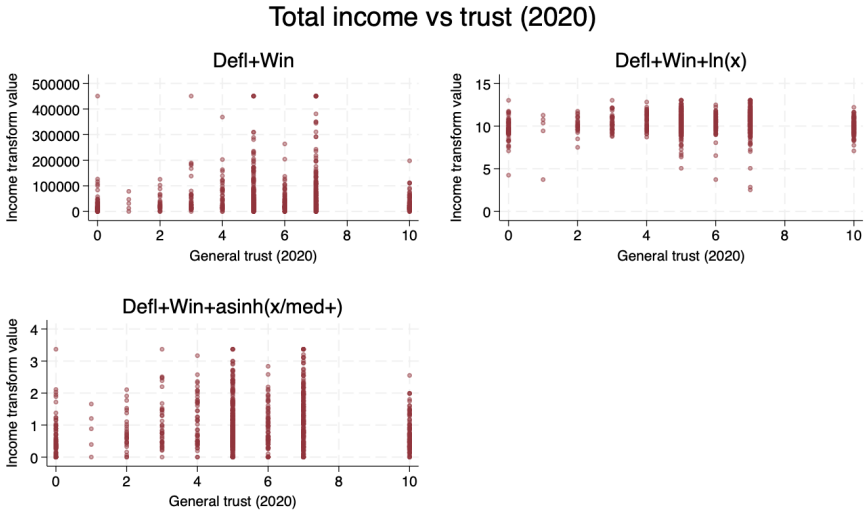



**Figure 28:** Total income measures vs. trust (2020)

### 3.6.2 Trust and returns

The scatterplots for trust and the measures of returns suggest stronger evidence for this hump shaped relationship. Especially for the smaller portfolio compositions (core, core and retirement), return values are highest in the middle and lower at the lowest and highest trust values.

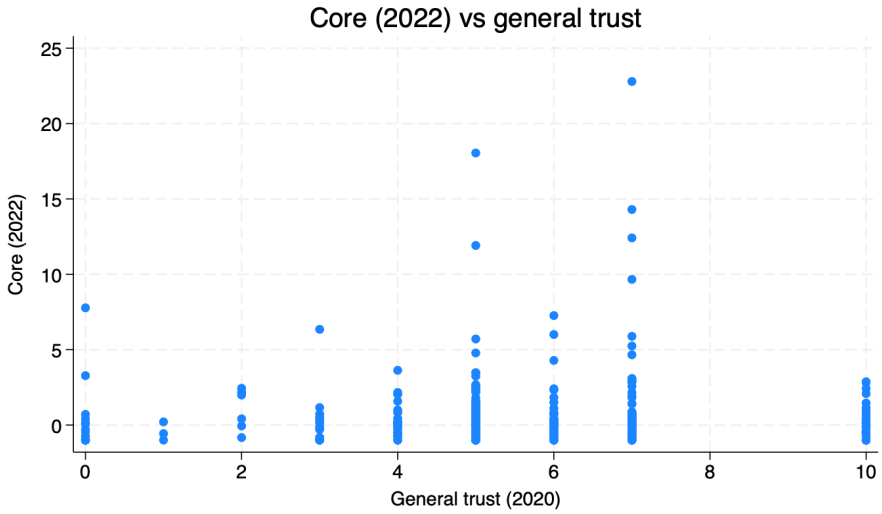



**Figure 29:** Returns to core assets vs. trust (2022)

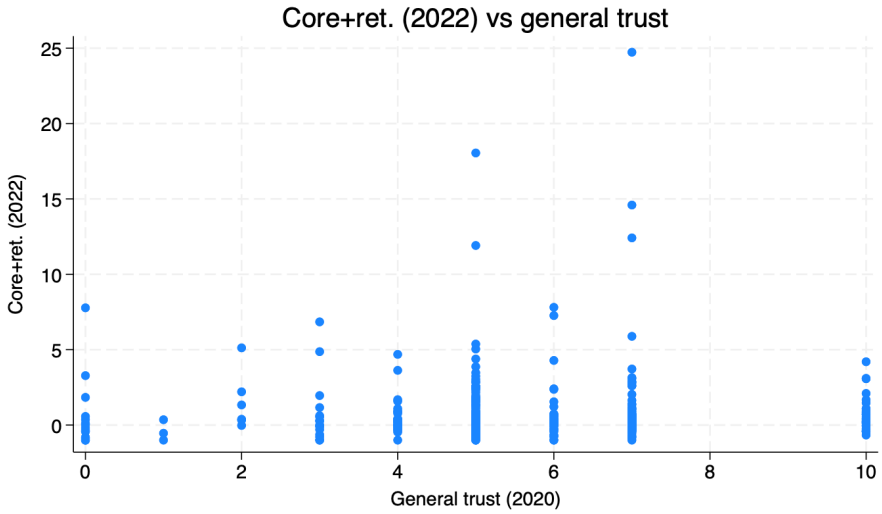



**Figure 30:** Returns to core and retirement assets vs. trust (2022)

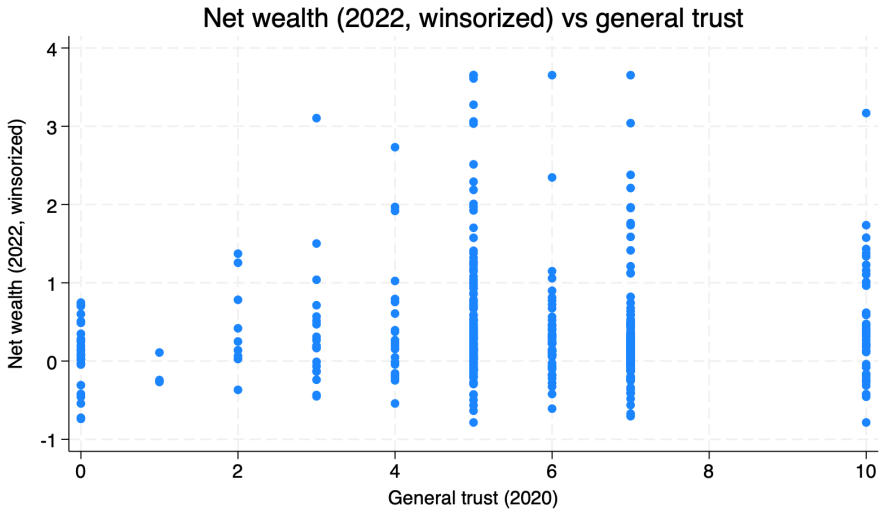



**Figure 31:** Returns to net wealth vs. trust (2022)

## 3.7 Demographics

As the HRS oversample older households, I take a quick look at demographic variables that will be important for the statistical analysis later.

*Panel A: Demographics (general)*

| Variable | N | Mean |
|---|---|---|
| Age | 15723 | 68.089 |
| Female | 15723 | 0.594 |
| Years of education | 15651 | 12.961 |
| Married | 15685 | 0.537 |
| Immigrant | 15718 | 0.172 |
| Born in U.S. | 15718 | 0.828 |
| Race: White (NH) | 8865 | 0.565 |
| Race: Black (NH) | 3381 | 0.216 |
| Race: Hispanic | 2678 | 0.171 |
| Race: Other (NH) | 765 | 0.049 |
| Working (in labor force) | 15479 | 0.386 |

*Panel B: Demographics (other controls)*

| Variable | N | Mean |
|---|---|---|
| Depression | 14998 | 1.55 |
| Health conditions | 15723 | 2.39 |
| Medicare | 15498 | 0.60 |
| Medicaid | 15383 | 0.14 |
| Life insurance | 15340 | 0.53 |
| Times divorced | 15723 | 0.57 |
| Times widowed | 15723 | 0.22 |

Mean = pct.

**Table 8:** Demographic summary statistics

The literature on trust already documents a signficant statistical relationship between trust and mental health. For that reason, I use the HRS RAND measure for a mental health index and show a clearly negative, linear relationship with trust.

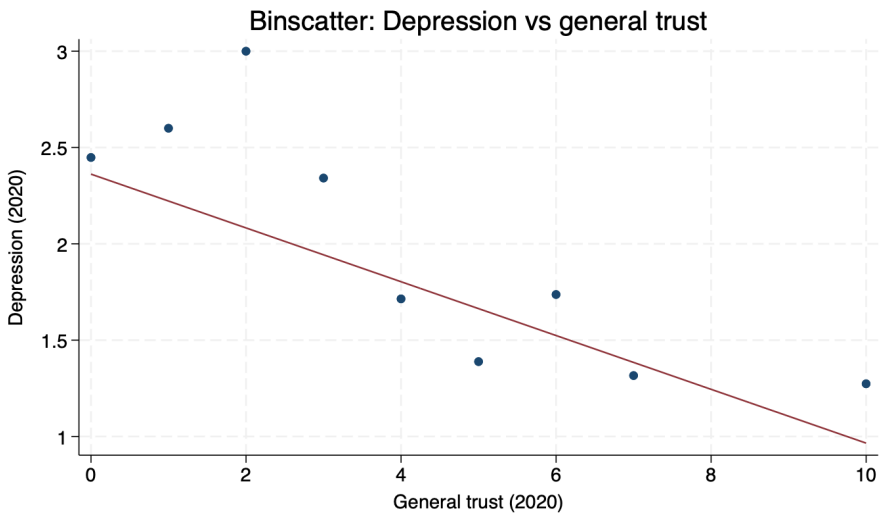



**Figure 32:** Depression vs. trust

# 4 Results

## 4.1 Trust and economic performance

After examining the determinants of trust, I focus the analysis of the empirical relationship between both income and returns and trust on a subset of measures. For income, I omit the analysis for labor income. I also exclude the analysis of the log of total income. Labor income results are not statistically significant. Given the oversampling of older, retired households, lack of predictive power for trust on labor income is not alarming. Total income results are robust to both log and IHS transformations.

For returns, I omit core and core and retirement portfolio results and instead focus the statistical analysis on returns to net wealth only. The results for the smaller portfolio compositions are largely insignificant across specifications (though the results are more encouraging when retirement assets are excluded). Again, with more retirees in the sample, an analysis of retirement assets may take more creative efforts. That said, I show significant that trust is predictive for the raw measure of returns to net wealth, and the relationship is robust to different trimming and winsorization choices.[8]

[8]This is important given the difficulty with computing returns in the U.S. due to lack of available data on flows in and out of narrowly defined asset classes.

### 4.1.1 Total income and trust

First I test whether the hump-shaped relationship between trust and income in Butler, Giuliano, and Guiso 2016 holds for the 2020 HRS cohort. In Table 4.1.1, we see the coefficients for both the linear and quadratic terms are significant; the predictive power survives even when the additional controls important for explaining income are included. Since the goal is to uncover whether there is a maximal level of trust regarding economic performance, then we care about the significance of both trust coefficients. Here, we reject the hypothesis that both coefficients are 0.[9]

**Table 9:** Total income (2020) on general trust

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Trust | 0.03*** | 0.19*** | 0.00 | 0.05** |
| | (0.01) | (0.02) | (0.01) | (0.02) |
| Trust$^2$ | | −0.02*** | | −0.00** |
| | | (0.00) | | (0.00) |
| Female | | | −0.27*** | −0.27*** |
| | | | (0.04) | (0.04) |
| Years of education | | | 0.07*** | 0.07*** |
| | | | (0.01) | (0.01) |
| Married | | | 0.14*** | 0.14*** |
| | | | (0.04) | (0.04) |
| Born in U.S. | | | 0.02 | 0.02 |
| | | | (0.07) | (0.07) |
| In labor force | | | 0.50*** | 0.49*** |
| | | | (0.05) | (0.05) |
| NH Black | | | −0.24*** | −0.23*** |
| | | | (0.05) | (0.05) |
| Hispanic | | | −0.24*** | −0.23*** |
| | | | (0.07) | (0.07) |
| NH Other | | | −0.27*** | −0.27*** |
| | | | (0.09) | (0.09) |
| Constant | 0.83*** | 0.53*** | −0.48*** | −0.51*** |
| | (0.05) | (0.06) | (0.14) | (0.16) |
| Observations | 900.00 | 900.00 | 890.00 | 890.00 |
| Adj. R-squared | 0.01 | 0.05 | 0.35 | 0.35 |
| Joint test: Trust p-value | . | 0.00 | . | 0.08 |
| Joint test: Age bins p-value | . | . | 0.00 | 0.00 |
| Joint test: Race p-value | . | . | 0.00 | 0.00 |

.

I also conduct the joint Wald test for the null hypotheses that the coefficients on the age bins and race categories are jointly equal to zero. In both cases, the null hypothesis is rejected at

[9] This rejection of the Wald test for the linear and quadratic trust terms (i.e. reject both coefficients are zero) will be true for most of the models analyzed in this paper.

conventional significance levels.

With income data available from 2002-2022, I consider the statistical relationship between average total income and trust. This can be seen in Table 4.1.1. Not only does the pattern persist, but the significance is stronger not only for trust coefficients, but for the other explanatory variables as well. Again, the Wald tests on the coefficients for race and age bins are rejected, meaning each matter for long-run total income.

**Table 10:** Average total income (avg defl wins, IHS) on General trust (2020)

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Trust | 0.03*** | 0.21*** | 0.00 | 0.05*** |
| | (0.01) | (0.02) | (0.01) | (0.02) |
| Trust$^2$ | | −0.02*** | | −0.00*** |
| | | (0.00) | | (0.00) |
| Female | | | −0.25*** | −0.25*** |
| | | | (0.04) | (0.04) |
| Years of education | | | 0.08*** | 0.08*** |
| | | | (0.01) | (0.01) |
| Married | | | 0.20*** | 0.19*** |
| | | | (0.04) | (0.04) |
| Born in U.S. | | | −0.03 | −0.03 |
| | | | (0.06) | (0.06) |
| In labor force | | | 0.37*** | 0.37*** |
| | | | (0.04) | (0.04) |
| NH Black | | | −0.27*** | −0.26*** |
| | | | (0.05) | (0.05) |
| Hispanic | | | −0.29*** | −0.28*** |
| | | | (0.06) | (0.06) |
| NH Other | | | −0.28*** | −0.29*** |
| | | | (0.08) | (0.08) |
| Constant | 0.90*** | 0.58*** | −0.62*** | −0.66*** |
| | (0.05) | (0.05) | (0.15) | (0.17) |
| Observations | 900.00 | 900.00 | 890.00 | 890.00 |
| Adj. R-squared | 0.01 | 0.07 | 0.42 | 0.43 |
| Joint test: Trust+Trust$^2$ p-value | . | 0.00 | . | 0.03 |
| Joint test: Age bins p-value | . | . | 0.00 | 0.00 |
| Joint test: Race p-value | . | . | 0.00 | 0.00 |

.

It is important to note that this is where the assumption regarding constant trust becomes required. For the previous model, all of the variables are contemporaneous. With average total income as the dependent variable, trust is viewed as a persistent trait about respondents that influences economic decisions over time so that it matters for long-run income. [10]

[10] Lesmeister, Limbach, and Goergen 2019 is an example of work which also assumes trust is time invariant.

### 4.1.2 Returns to net wealth and trust

The presence of hump-shaped empirical relationship between earnings and trust is important because it aligns with previous literature Butler, Giuliano, and Guiso 2016. Given the oversampling of older and retired households by the HRS, this is an important first step towards using the dataset to argue a similar pattern trust and returns.

For earnings and trust, I used contemporaneous measures to see if the HRS results aligned with the literature. Since returns to net wealth are of primary interest, and the inherent struture of returns and financial contracts is about future payoff/repayment, I use the 2020 trust measure to predict returns in 2022 (the following wave of the survey).

Table 4.1.2 shows the relationship between 2022 returns to net worth and 2020 trust and other control variables. These cross-section results are discouraging in that, the significance of the trust coefficients erodes when control variables are included. The Wald test on the linear and quadratic term for trust also fails to reject the null hypothesis that the coefficients are both zero.

Among the controls which are significant for returns to net wealth, education is positive and has the strongest predictive power. Race and gender matter here, where all racial minority groups and women take a hit to returns for the cross section. I also conduct Wald tests for the null hypothesis that the coefficients on the age bins, wealth deciles, and race categories are jointly equal to zero. In all three cases, the null hypothesis is rejected at conventional significance levels, indicating that these groups of variables contribute significantly to explaining variation in the returns to net wealth.

**Table 11:** 2022 Returns to net wealth (2022) on General trust (2020) (5 percent winsorized)

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Trust | 0.01 | 0.06** | 0.01 | 0.03 |
| | (0.01) | (0.02) | (0.01) | (0.03) |
| Trust$^2$ | | −0.00* | | −0.00 |
| | | (0.00) | | (0.00) |
| Female | | | −0.10** | −0.09** |
| | | | (0.04) | (0.04) |
| Years of education | | | 0.03*** | 0.02*** |
| | | | (0.01) | (0.01) |
| Married | | | 0.08 | 0.08 |
| | | | (0.05) | (0.05) |
| Born in U.S. | | | 0.03 | 0.03 |
| | | | (0.05) | (0.05) |
| NH Black | | | −0.16** | −0.15** |
| | | | (0.06) | (0.07) |
| Hispanic | | | −0.13* | −0.13* |
| | | | (0.07) | (0.07) |
| NH Other | | | −0.05 | −0.05 |
| | | | (0.12) | (0.12) |
| In labor force | | | 0.13** | 0.13** |
| | | | (0.05) | (0.05) |
| _cons | 0.24*** | 0.13** | −0.49** | −0.47** |
| | (0.06) | (0.07) | (0.20) | (0.20) |
| Observations | 497.00 | 497.00 | 406.00 | 406.00 |
| Adj. R-squared | 0.00 | 0.01 | 0.23 | 0.23 |
| Joint test: Trust p-value | | 0.03 | | 0.54 |
| Joint test: Age bins p-value | . | . | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | . | . | 0.00 | 0.00 |
| Joint test: Race p-value | . | . | 0.06 | 0.07 |

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The main empirical object of analysis for this paper is the distribution of returns to net worth for respondents in the HRS from 2002 to 2022. The persistent component of returns is of particular interest to the heterogeneous agent literature, and the panel regression version of the model with individual fixed effects will allow us to describe this components. Evidence of individuals persistently earning different returns on average is motivation for allowing households to be heterogeneous in their returns in the macro modeling framework, which I explore in another paper.

I also inspect the relationship between average annual returns over the period and trust in Table 4.1.2. Here the hump-shaped relationship between trust and returns has empirical support. The individual coefficients are significant, and the Wald test for both the linear and quadratic term suggests that the model with both terms included does better at explaining returns than the model without (when both coefficients are zero). Labor force participation is highly significant and positive for average returns.

Among the controls significant for returns to net wealth, education remains a strong predictor, while gender and race also show notable effects in the analysis over average returns for the period. I conduct Wald tests and find that the null hypothesis is rejected for the coefficients of trust and age bins being jointly equal to zero.

**Table 12:** Average Returns to net wealth (avg) on General trust (2020) (5 percent winsorized)

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Trust | 0.01** | 0.07*** | 0.01** | 0.04*** |
| | (0.00) | (0.01) | (0.01) | (0.02) |
| $\text{Trust}^2$ | | −0.01*** | | −0.00** |
| | | (0.00) | | (0.00) |
| Female | | | −0.04* | −0.04* |
| | | | (0.02) | (0.02) |
| Years of education | | | 0.01 | 0.00 |
| | | | (0.00) | (0.00) |
| Married | | | −0.00 | −0.00 |
| | | | (0.03) | (0.03) |
| Born in U.S. | | | −0.02 | −0.02 |
| | | | (0.04) | (0.04) |
| NH Black | | | 0.02 | 0.02 |
| | | | (0.04) | (0.04) |
| Hispanic | | | −0.10** | −0.10** |
| | | | (0.05) | (0.05) |
| NH Other | | | −0.03 | −0.03 |
| | | | (0.08) | (0.08) |
| In labor force | | | 0.10*** | 0.10*** |
| | | | (0.03) | (0.03) |
| Midwest | | | −0.01 | −0.01 |
| | | | (0.04) | (0.04) |
| South | | | 0.03 | 0.03 |
| | | | (0.04) | (0.04) |
| West | | | 0.14*** | 0.14*** |
| | | | (0.04) | (0.04) |
| _cons | 0.15*** | 0.04 | −0.04 | −0.02 |
| | (0.03) | (0.04) | (0.10) | (0.10) |
| Observations | 750.00 | 750.00 | 592.00 | 592.00 |
| Adj. R-squared | 0.01 | 0.03 | 0.08 | 0.09 |
| Joint test: Trust p-value | . | 0.00 | . | 0.01 |
| Joint test: Age bins p-value | . | . | 0.02 | 0.11 |
| Joint test: Wealth deciles p-value | . | . | 0.09 | 0.10 |
| Joint test: Race p-value | . | . | 0.12 | 0.11 |
| Joint test: Region p-value | . | . | 0.00 | 0.00 |

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## 4.2 Returns from 2002-2022

The main empirical analysis of this paper involves computing returns across several years and describing the persistent component of returns. As my motivation is characterizing heterogeneity in the rate of return across individuals, I use model specifcations similar to those in Fagereng et al. 2020. First, a pooled OLS regression with a baseline set of controls. Second, a similar pooled regression aimed at controlling for risk exposure. Each of these specifications are captupred by the linear panel data regression model

$$r_{i,t}^{net} = \alpha + X_{i,t}'\beta + \varepsilon_{i,t}$$

where $X_{i,t}$ the set of regressors includes both time-varying and time-invariant demographic variables, as well as the trust measure as linear or with both linear and quadratic terms present.

The third specification has two stages. I begin with a panel regression including individual and year fixed effects along with the time-invariant regressors from the second model. This regression model is extended so that the error term $\varepsilon_{i,t}$ can be written as

$$\varepsilon_{i,t} = f_{i,t} + e_{i,t},$$

where $e_{i,t}$ may possibly be serially correlated. Then I regress the estimated individual fixed effect onto the time invariant regressors

$$\hat{f}_{i,t} = \bar{X}_{i,t} + u_{i,t},$$

where $\bar{X}_{i,t}$ is the set of time-invariant regressor (subset of the full set of controls). This will give us a sense of how the persistent component changes with the variables excluded from the first stage.

### 4.2.1 The pooled regression with constant trust

In Table 4.2.1, I show three versions of the model: no trust term, linear trust term, both linear and quadratic terms. The time-invariant trust measure displays the hump-shaped relationship for returns to net wealth in terms inference on the estimated coefficients and the joint Wald test.

**Table 13:** Panel: returns to net wealth (5 percent winsorized)

| | (1) | (2) | (3) |
|---|---|---|---|
| Years of education | $-0.00^{***}$ | $-0.01^{***}$ | $-0.01^{***}$ |
| | (0.00) | (0.00) | (0.00) |
| Female | $-0.02^{***}$ | $-0.03^{*}$ | $-0.03^{*}$ |
| | (0.00) | (0.02) | (0.02) |
| NH Black | $0.06^{***}$ | $0.05^{*}$ | $0.05^{*}$ |
| | (0.00) | (0.03) | (0.03) |
| Hispanic | $0.03^{***}$ | $-0.01$ | $-0.01$ |
| | (0.01) | (0.03) | (0.03) |
| NH Other | $0.02^{**}$ | 0.01 | 0.01 |
| | (0.01) | (0.04) | (0.04) |
| In labor force | $0.03^{***}$ | $0.03^{*}$ | $0.03^{*}$ |
| | (0.00) | (0.02) | (0.02) |
| Married | $-0.06^{***}$ | $-0.06^{***}$ | $-0.06^{***}$ |
| | (0.00) | (0.02) | (0.02) |
| Born in U.S. | $0.02^{***}$ | 0.03 | 0.03 |
| | (0.00) | (0.02) | (0.02) |
| Trust | | $0.01^{**}$ | $0.03^{***}$ |
| | | (0.00) | (0.01) |
| $\text{Trust}^2$ | | | $-0.00^{**}$ |
| | | | (0.00) |
| _cons | 0.15 | $-0.56^{***}$ | $-0.54^{***}$ |
| | (0.12) | (0.06) | (0.06) |
| Observations | 102091.00 | 2899.00 | 2899.00 |
| Adj. R-squared | 0.03 | 0.04 | 0.04 |
| Joint test: Trust p-value | . | . | 0.00 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.00 | 0.01 | 0.01 |
| Joint test: Race p-value | 0.00 | 0.26 | 0.20 |
| Joint test: Region p-value | 0.00 | 0.09 | 0.11 |
| Joint test: Year p-value | 0.00 | 0.00 | 0.00 |

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Education and marital status also have strong predictive power in the panel setting. Along with the general trust meausre, the Wald tests are rejected for the coefficients on age bins, wealth deciles, and year fixed effects being jointly equal to zero.

### 4.2.2 Controlling for risk exposure

The key point about the documented persistent component to returns is that, even when controlling for risk preferences and exposure, individuals systematic earn different returns on the same assets[11].

[11] This point is further corroborated by evidence of heterogeneity in returns on safe assets like deposit accounts. An example from the academic literature is Deuflhard, Georgarakos, and Inderst 2014, but one can compare rates

Moreover, the descriptive statistics for mean returns show that returns do not seem stable over the time period (ex. Global Financial Crisis). To account for risk exposure in this environment, I add full interaction terms between shares invested in given asset classes (core, retirement, residential) and year. The regression results for the return measures are given in Table 4.2.2.

offered on deposit account at commercial banks like Bank of America versus local banks and see similar discrepancies in everyday life.

**Table 14:** Panel Spec 2: returns to net wealth (5 percent winsorized, share x year)

| | (1) | (2) | (3) |
|---|---|---|---|
| Years of education | $-0.01^{***}$ | $-0.01^{***}$ | $-0.01^{***}$ |
| | (0.00) | (0.00) | (0.00) |
| Female | $-0.02^{***}$ | $-0.02$ | $-0.02$ |
| | (0.00) | (0.02) | (0.01) |
| NH Black | $0.07^{***}$ | 0.04 | 0.04 |
| | (0.00) | (0.03) | (0.03) |
| Hispanic | $0.05^{***}$ | $-0.01$ | $-0.01$ |
| | (0.01) | (0.03) | (0.03) |
| NH Other | $0.02^{***}$ | 0.03 | 0.03 |
| | (0.01) | (0.04) | (0.04) |
| In labor force | $0.03^{***}$ | 0.02 | 0.02 |
| | (0.00) | (0.02) | (0.02) |
| Married | $-0.06^{***}$ | $-0.07^{***}$ | $-0.07^{***}$ |
| | (0.00) | (0.02) | (0.02) |
| Born in U.S. | $0.01^{*}$ | 0.02 | 0.02 |
| | (0.00) | (0.02) | (0.02) |
| Share core | $-0.06^{**}$ | 0.25 | 0.25 |
| | (0.03) | (0.24) | (0.24) |
| Share IRA | $-0.24^{***}$ | $-0.08$ | $-0.08$ |
| | (0.03) | (0.24) | (0.24) |
| Share residential | $-0.13^{***}$ | $-0.00$ | 0.01 |
| | (0.03) | (0.23) | (0.23) |
| Share long-term debt | $-0.02^{***}$ | $-0.15$ | $-0.16$ |
| | (0.01) | (0.27) | (0.27) |
| Share other debt | $-0.01$ | 0.13 | 0.14 |
| | (0.01) | (0.17) | (0.16) |
| Trust | | $0.01^{**}$ | $0.03^{***}$ |
| | | (0.00) | (0.01) |
| Trust$^2$ | | | $-0.00^{**}$ |
| | | | (0.00) |
| _cons | 0.17 | $-0.46^{**}$ | $-0.45^{**}$ |
| | (0.16) | (0.22) | (0.22) |
| Observations | 100837.00 | 2866.00 | 2866.00 |
| Adj. R-squared | 0.05 | 0.06 | 0.06 |
| Joint test: Trust+Trust$^2$ p-value | . | . | 0.01 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.00 | 0.03 | 0.02 |
| Joint test: Race p-value | 0.00 | 0.40 | 0.31 |
| Joint test: Region p-value | 0.00 | 0.05 | 0.06 |
| Joint test: Year p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Share (core, IRA, res) p-value | 0.00 | 0.28 | 0.29 |
| Joint test: Share (debt long, other) p-value | 0.00 | 0.64 | 0.58 |
| Joint test: Share core x year p-value | 0.00 | 0.01 | 0.01 |
| Joint test: Share IRA x year p-value | 0.00 | 0.01 | 0.01 |
| Joint test: Share res x year p-value | 0.00 | 0.01 | 0.01 |
| Joint test: Share debt long x year p-value | 0.00 | 0.13 | 0.13 |
| Joint test: Share debt other x year p-value | 0.61 | 0.00 | 0.00 |

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The predictive power of the trust coefficents survives after controlling for risk. Although the uninteracted terms for shares in various asset classes are not signficant, the performance for all asset classes and for the "Other debt" liability regarding the Wald test suggests that portfolio allocation among these assets and debt over the time period is explanatory for the variation in returns.

Education and marital status remain highly predictive as well even after controlling for risk exposure. Along with the general trust measure, the Wald tests are rejected for the coefficients on age bins, wealth deciles, region, and year fixed effects being jointly equal to zero. The interaction terms between asset class shares and year are also jointly significant for all asset classes and for the "Other debt" liabilities class.

#### 4.2.3 Fixed effects and the persistent component of returns

Evidence of a maximal level of trust associated with returns to net wealth is a novel extension of the findings by Butler, Giuliano, and Guiso 2016. I now turn attention to estimating the persistent component of returns by estimated the model with individual fixed effects. Notably, inclusion of this term will require we drop the time-invariant regressors from the set of control variables.

In this setting, the coefficents on the uninteracted share terms for all of the available asset (core, retirement, residential) and liabilities (long, other) are now significant, which can be seen in Table 4.2.3.

**Table 15:** Panel Spec 3: returns to net wealth (5 percent winsorized, share x year)

| | (1) |
|---|---|
| In labor force | 0.03*** |
| | (0.00) |
| Married | −0.05*** |
| | (0.01) |
| Share core | 0.25*** |
| | (0.04) |
| Share IRA | 0.08** |
| | (0.04) |
| Share residential | 0.14*** |
| | (0.04) |
| Share long-term debt | −0.01* |
| | (0.01) |
| Share other debt | −0.08** |
| | (0.04) |
| _cons | −0.49*** |
| | (0.14) |
| Observations | 101088.00 |
| Within $R^2$ | 0.19 |
| $\rho$ | 0.66 |
| $\sigma_u$ | 0.46 |
| $\sigma_e$ | 0.33 |
| Joint test: Age bins p-value | 0.00 |
| Joint test: Region p-value | 0.20 |
| Joint test: Wealth deciles p-value | 0.00 |
| Joint test: Year p-value | 0.00 |
| Joint test: Share (core, IRA, res) p-value | 0.00 |
| Joint test: Share (debt long, other) p-value | 0.01 |

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Employment status and marital status are significant predictors of returns to net wealth when fixed effects are included. Wald tests reject the null hypothesis that the coefficients on age bins, wealth deciles, and year fixed effects are jointly equal to zero, indicating their relevance in explaining within-individual return variation. The asset share variables (core, IRA, residential) and debt shares (long-term and other) are also jointly significant, confirming the importance of portfolio composition even after controlling for individual fixed effects.

In terms of the estimated fixed effect, Table 4.2.3 shows that a large share of total variance is captured by $\rho$, which is evidence in favor of heterogeneity in returns across respondents. Consistent with this, $\sigma_u$ (the standard deviation of person-specific intercepts) is sizable, suggesting substantial persistent differences in return levels across individuals.

By contrast, $\sigma_e$ captures the remaining within-person time variation in returns that is not explained by observed controls. The within-$R^2$ is therefore interpreted as the share of within-respondent return movements explained by the included time-varying covariates – about only a third

of the within-person variation in returns is left after controlling for observables, and importantly, individual fixed effects.

The distribution of estimated individual fixed effects (demeaned, winsorized top/bottom 1%) from this model is shown in the following figure **??**. The is centered close to zero, which indicates that most households have only modest persistent deviations from the return predicted by observed covariates and year effects. Relative to the Norwegian benchmark, the HRS distribution is visibly more compressed and right-skewed: there is a longer positive tail and fewer large negative fixed effects. This pattern is consistent with a small group of households exhibiting persistently higher net-wealth returns, while most households cluster around near-zero persistent effects.

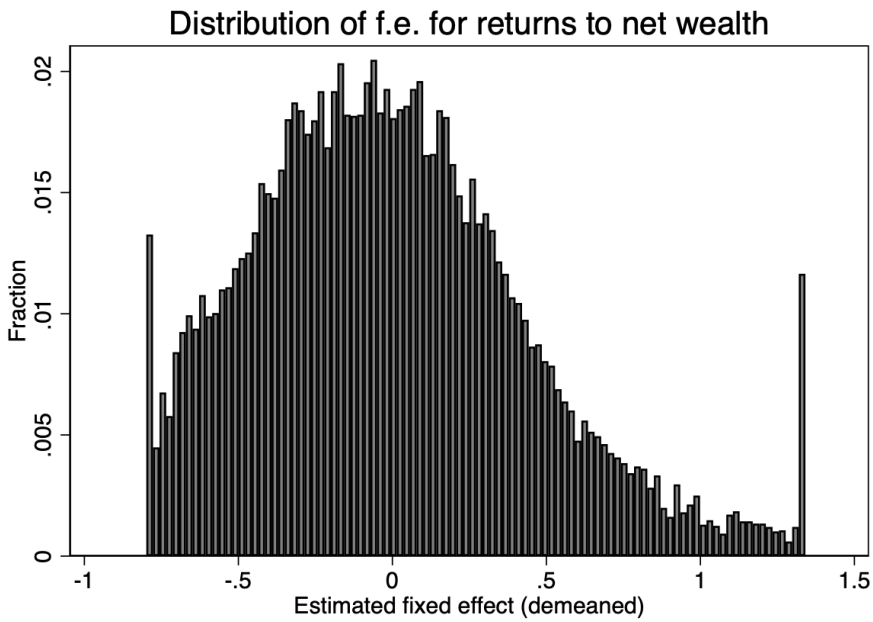


**Figure 33:** FE distribution: returns to net wealth

In the second stage, I predict the estimated fixed effects using the time-invariantregressors (education, gender, race, born in U.S., trust). The second-stage results for winsorized returns to net wealth are given in Table 4.2.3.

**Table 16:** Second-stage: FE from returns to net wealth (5 percent winsorized) on time-invariant vars

| | (1) | (2) | (3) |
|---|---|---|---|
| Years of education | $-0.04^{***}$ | $-0.04^{***}$ | $-0.04^{***}$ |
| | (0.00) | (0.01) | (0.01) |
| Female | $0.01^{**}$ | 0.04 | 0.04 |
| | (0.01) | (0.04) | (0.04) |
| NH Black | $0.30^{***}$ | $0.25^{***}$ | $0.25^{***}$ |
| | (0.01) | (0.05) | (0.05) |
| Hispanic | $0.22^{***}$ | $0.23^{***}$ | $0.23^{***}$ |
| | (0.01) | (0.05) | (0.05) |
| NH Other | $0.16^{***}$ | 0.08 | 0.09 |
| | (0.02) | (0.07) | (0.07) |
| Born in U.S. | $0.07^{***}$ | $0.14^{***}$ | $0.14^{***}$ |
| | (0.01) | (0.05) | (0.05) |
| Trust | | $-0.00$ | 0.02 |
| | | (0.01) | (0.02) |
| $\text{Trust}^2$ | | | $-0.00$ |
| | | | (0.00) |
| _cons | $0.39^{***}$ | $0.40^{***}$ | $0.37^{***}$ |
| | (0.02) | (0.10) | (0.10) |
| Observations | 21543.00 | 558.00 | 558.00 |
| Adj. R-squared | 0.14 | 0.16 | 0.16 |
| Joint test: Trust p-value | . | . | 0.65 |
| Joint test: Race p-value | 0.00 | 0.00 | 0.00 |

.

The estimated individual fixed effect is the average return for an individual after removing what part of average returns is predicted by the time-varying regressors. Race and education are predictive for the fixed effect, but unfortunately the trust variable is no longer significant for any version of the model.

Although the trust coefficients fail to reject the test that they are equal to 0, I present a figure 34 of the relationship between estimated fixed effects and trust here. Withstanding some observations, it does seem that the maximal levels of the estimated fixed effect are associated with the medium range of trust values between 0 and 10.

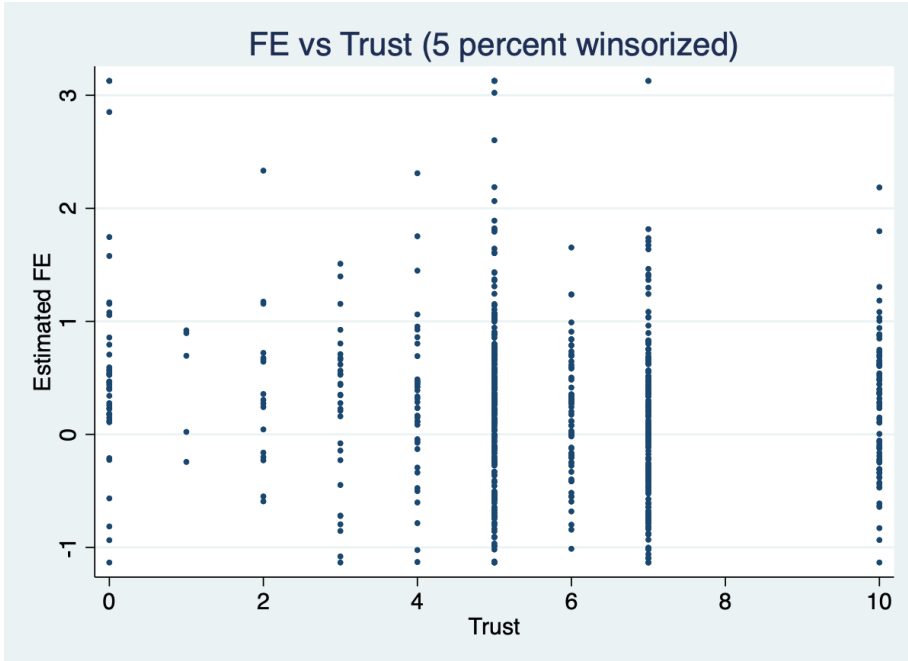


**Figure 34:** FE vs. trust (5 percent winsorized)

## 4.3 The "right amount of trust" in the HRS

The model specifications including both a linear and quadratic term for trust are of the form

$$r_{i,t}^{net} = \alpha + \beta_1 \mathit{Trust}_i + \beta_2 \mathit{Trust}_i^2 + X_i'\gamma + \varepsilon_i$$

. Taking the partial derivative with respect to trust is

$$\frac{\partial r_{i,t}^{net}}{\partial \mathit{Trust}_i} = \beta_1 + 2\beta_2 \mathit{Trust}_i$$

. Setting this to zero yields the critical point $\mathit{Trust}^* = \frac{-\beta_1}{2\beta_2}$. When $\hat{\beta}_1 \geq 0, \hat{\beta}_2 \leq 0$ the relationship is hump-shaped (concave) and the estimated level of trust which maximizes returns to net wealth is

$$\mathit{Trust}^* = \frac{-\hat{\beta}_1}{2\hat{\beta}_2}$$

.

### 4.3.1 The maximal level of trust for income and returns

Table 4.3.1 reports the turning point $\mathit{Trust}^*$ for each specification, along with $\hat{\beta}_1$, $\hat{\beta}_2$, whether both coefficients are individually significant, and the joint test p-value.

**Table 17:** Estimated trust *Trust*$^*$ associated with maximal economic performance

| Model | N | $\hat{\beta}_1$ | $\hat{\beta}_2$ | Both sig.? | Joint test | (*Trust*)$^*$ |
|---|---|---|---|---|---|---|
| Total Income (2022) | 890 | 0.0494 | -0.0044 | Yes | 0.0766 | 5.6209 |
| Total Income (Avg.) | 890 | 0.0494 | -0.0046 | Yes | 0.0278 | 5.3329 |
| Returns to net wealth (2022) | 406 | 0.0309 | -0.0022 | No | 0.5431 | 6.8606 |
| Returns to net wealth (Avg.) | 593 | 0.0469 | -0.0034 | Yes | 0.0133 | 6.9121 |
| Panel specification 1 | 2,899 | 0.0279 | -0.0019 | Yes | 0.0018 | 7.4222 |
| Panel specification 2 | 2,866 | 0.0264 | -0.0018 | Yes | 0.0073 | 7.3211 |

Across all six specifications, $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$, so the relationship is hump-shaped and *Trust*$^*$ is a maximum. Both trust coefficients are individually significant in five of the six specifications: total income in 2022, average total income, average returns to net worth, and returns to net worth in panel specifications 1 and 2. The Wald test for the linear and quadratic trust terms is rejected at the 5% level in average total income, average returns to net worth, and both panel specifications. Estimated trust *Trust*$^*$ associated with maximal economic performance ranges from about 5.3 for average total income to 7.4 for returns to net worth in the baseline panel specification, and in every case it lies within the observed trust scale $[0, 10]$.

### 4.3.2 Delta method for estimated trustassociated with maximal economic performance

Because *Trust*$^*$ is a nonlinear function of $\hat{\beta}_1$ and $\hat{\beta}_2$, Stata uses the delta method to approximate its standard error. Table 4.3.2 reports *Trust*$^*$, SE(*Trust*$^*$), 95% confidence intervals, and p-values which test $H_0 : \textit{Trust}^* = 0$.

**Table 18:** Delta-method results for trust turning points

| Model | N | $(Trust)^*$ | SE$(Trust)^*$ | p$(Trust)^*$ | $\underline{CI}$ | $\overline{CI}$ |
|---|---|---|---|---|---|---|
| Total Income (2022) | 890 | 5.6209 | 0.8174 | 0.0000 | 4.0188 | 7.2229 |
| Total Income (Avg.) | 890 | 5.3329 | 0.6858 | 0.0000 | 3.9888 | 6.6770 |
| Returns to net wealth (2022) | 406 | 6.8606 | 2.8117 | 0.0147 | 1.3498 | 12.3714 |
| Returns to net wealth (Avg.) | 593 | 6.9121 | 0.9832 | 0.0000 | 4.9850 | 8.8392 |
| Panel spec. 1 | 2,899 | 7.4222 | 1.0247 | 0.0000 | 5.4138 | 9.4306 |
| Panel spec. 2 | 2,866 | 7.3211 | 1.0468 | 0.0000 | 5.2693 | 9.3728 |

# 5 2SLS approach for trust and returns

## 5.1 First stage: predict trust

Table 5.1 describes the set of controls and determinants of general trust. Column (1) includes demographics only: gender, education, marital status, and race/ethnicity. Column (2) adds population size and the "Potential for Deception" measures for social security, medicare/medicaid, banks, financial advisors, mutual funds, and insurance companies. Column (3) adds health conditions, insurance coverage, life insurance, and marital history. As Alesina and Ferrara 2000 suggests, demographic variables tend to determine trust. In the HRS data, the estimated coefficients for non-Hispanic Black and for marital status have the strongest significance; the race coefficient is negative, consistent with Alsan and Wanamaker 2018. Depression is negative and significant across specifications.

**Table 19:** Determinants of general trust

| | (1) | (2) | (3) |
|---|---|---|---|
| Female | 0.26 | 0.60*** | 0.61*** |
| | (0.17) | (0.22) | (0.23) |
| Years of education | 0.04 | 0.00 | −0.00 |
| | (0.03) | (0.05) | (0.05) |
| Married | 0.38** | −0.01 | −0.13 |
| | (0.18) | (0.24) | (0.26) |
| NH Black | −1.06*** | −1.19*** | −1.25*** |
| | (0.22) | (0.31) | (0.31) |
| Hispanic | −0.43 | −0.38 | −0.39 |
| | (0.27) | (0.40) | (0.40) |
| NH Other | −0.17 | −1.00 | −1.26* |
| | (0.31) | (0.63) | (0.64) |
| Depression | −0.17*** | −0.14** | −0.12* |
| | (0.04) | (0.06) | (0.06) |
| Small/med city (10k-100k) | | 0.00 | −0.10 |
| | | (0.29) | (0.29) |
| Large metro (100k+) | | 0.10 | 0.08 |
| | | (0.29) | (0.29) |
| Deception: Social Security | | −0.00 | 0.00 |
| | | (0.00) | (0.00) |
| Deception: Medicare/Medicaid | | 0.00 | 0.00 |
| | | (0.00) | (0.00) |
| Deception: Banks | | −0.00 | −0.00* |
| | | (0.00) | (0.00) |
| Deception: Financial advisors | | −0.00 | −0.00 |
| | | (0.00) | (0.00) |
| Deception: Mutual funds | | 0.00 | 0.00 |
| | | (0.00) | (0.00) |
| Deception: Insurance companies | | −0.00 | −0.00* |
| | | (0.00) | (0.00) |
| Health conditions | | | −0.06 |
| | | | (0.09) |
| Covered by Medicare | | | −0.01 |
| | | | (0.35) |
| Covered by Medicaid | | | −0.25 |
| | | | (0.37) |
| Has life insurance | | | 0.39* |
| | | | (0.23) |
| Number of reported divorces | | | −0.07 |
| | | | (0.13) |
| Number of reported times being widowed | | | −0.11 |
| | | | (0.32) |
| Constant | 2.83 | 1.02 | 1.54 |
| | (1.83) | (0.83) | (0.94) |
| Observations | 894.00 | 502.00 | 493.00 |
| Adj. R-squared | 0.11 | 0.14 | 0.15 |
| Joint test: Deception vars p-value | . | 0.62 | 0.40 |

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### 5.1.1 Relevance

Table 5.1.1 shows the first-stage trust models with depression instrument variants used in the IV checks. The coefficient on contemporaneous depression is negative and significant in the controlled specification; the 2018 lag is also negative and significant when entered alone. When both 2016 and 2018 lags are included, only the 2018 lag remains significant. The "ever depressed" indicator is negative and significant. This pattern supports instrument relevance.

| | No lag | One lag | Two lags | Ever depressed |
|---|---|---|---|---|
| Female | 0.28 | 0.38* | 0.42** | 0.32 |
| | (0.21) | (0.21) | (0.21) | (0.21) |
| Years of education | 0.04 | 0.03 | 0.04 | 0.03 |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| Married | 0.09 | 0.13 | 0.11 | 0.12 |
| | (0.22) | (0.22) | (0.23) | (0.22) |
| NH Black | −1.13*** | −1.16*** | −1.16*** | −1.06*** |
| | (0.27) | (0.28) | (0.28) | (0.27) |
| Hispanic | −0.18 | −0.04 | 0.01 | −0.21 |
| | (0.37) | (0.39) | (0.39) | (0.36) |
| NH Other | −0.75 | −0.91 | −0.91 | −0.70 |
| | (0.54) | (0.59) | (0.60) | (0.54) |
| In labor force | 0.02 | 0.01 | 0.02 | 0.04 |
| | (0.21) | (0.22) | (0.22) | (0.21) |
| Midwest | 0.42 | 0.52 | 0.52 | 0.47 |
| | (0.32) | (0.33) | (0.34) | (0.31) |
| South | 0.27 | 0.47 | 0.48 | 0.31 |
| | (0.29) | (0.31) | (0.32) | (0.29) |
| West | 0.35 | 0.43 | 0.45 | 0.40 |
| | (0.31) | (0.34) | (0.35) | (0.31) |
| Born in U.S. | 0.62** | 0.60* | 0.58* | 0.60* |
| | (0.31) | (0.33) | (0.34) | (0.31) |
| Depression (no lag) | −0.11** | | | |
| | (0.05) | | | |
| Depression (lag 1) | | −0.19*** | −0.17** | |
| | | (0.05) | (0.08) | |
| Depression (lag 2) | | | −0.04 | |
| | | | (0.08) | |
| Ever depressed | | | | −0.79*** |
| | | | | (0.25) |
| Constant | −0.01 | −0.32 | −0.33 | 0.70 |
| | (0.73) | (0.75) | (0.76) | (0.78) |
| Observations | 663.00 | 616.00 | 607.00 | 663.00 |
| Adj. R-squared | 0.12 | 0.14 | 0.14 | 0.13 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.02 | 0.02 | 0.03 | 0.01 |
| Joint test: Race p-value | 0.00 | 0.00 | 0.00 | 0.00 |
| Joint test: Region p-value | 0.57 | 0.40 | 0.42 | 0.47 |

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The first-stage relevance checks in table 5.1.1 show depression terms consistently negative. I focus on the set of variables $Z = \{\texttt{Depression}, \texttt{Population}, \texttt{Deception}_l\}$, where $l$ indexes the "Potential for Deception" measures in the HRS, as potential instruments. The remaining control variables match the reduced-form specifications. Because depression is negative and strongly predictive of trust in the first-stage, the core IV exercises focus on depression-based instruments.

### 5.1.2 Exclusion restriction

For the 2022 cross section, table 5.1.2 regresses returns to net wealth on controls, trust, trust-squared, and depression instrument variants (no lag, one lag, two lags, ever depressed).

**Table 20:** Exclusion restriction: r5 cross-section (2022) on controls + depression + Trust + Trust$^2$

| | No lag | One lag | Two lags | Ever depressed |
|---|---|---|---|---|
| Female | $-0.09^{**}$ | $-0.07$ | $-0.07$ | $-0.10^{**}$ |
| | (0.04) | (0.04) | (0.05) | (0.04) |
| Years of education | $0.02^{***}$ | $0.02^{**}$ | $0.02^{**}$ | $0.02^{***}$ |
| | (0.01) | (0.01) | (0.01) | (0.01) |
| Married | 0.08 | 0.08 | 0.08 | $0.09^{*}$ |
| | (0.05) | (0.05) | (0.05) | (0.05) |
| NH Black | $-0.17^{**}$ | $-0.17^{**}$ | $-0.18^{***}$ | $-0.17^{**}$ |
| | (0.07) | (0.07) | (0.07) | (0.07) |
| Hispanic | $-0.20^{***}$ | $-0.23^{***}$ | $-0.24^{***}$ | $-0.19^{***}$ |
| | (0.07) | (0.07) | (0.07) | (0.07) |
| NH Other | $-0.12$ | $-0.15$ | $-0.15$ | $-0.13$ |
| | (0.13) | (0.14) | (0.14) | (0.13) |
| In labor force | $0.13^{***}$ | $0.13^{**}$ | $0.12^{**}$ | $0.14^{***}$ |
| | (0.05) | (0.05) | (0.05) | (0.05) |
| Midwest | $-0.06$ | $-0.08$ | $-0.07$ | $-0.06$ |
| | (0.07) | (0.07) | (0.08) | (0.07) |
| South | 0.05 | 0.01 | 0.02 | 0.05 |
| | (0.06) | (0.06) | (0.07) | (0.06) |
| West | $0.12^{*}$ | 0.12 | $0.14^{*}$ | 0.11 |
| | (0.07) | (0.08) | (0.08) | (0.07) |
| Depression (no lag) | 0.00 | | | |
| | (0.01) | | | |
| Depression (lag 1) | | $-0.04^{***}$ | $-0.03^{*}$ | |
| | | (0.01) | (0.02) | |
| Depression (lag 2) | | | $-0.02$ | |
| | | | (0.02) | |
| Ever depressed | | | | 0.06 |
| | | | | (0.05) |
| Trust | 0.03 | 0.04 | 0.03 | 0.04 |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| Trust$^2$ | $-0.00$ | $-0.00$ | $-0.00$ | $-0.00$ |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| Constant | $-0.53^{**}$ | $-0.42^{**}$ | $-0.44^{**}$ | $-0.61^{***}$ |
| | (0.21) | (0.21) | (0.22) | (0.22) |
| Observations | 405.00 | 384.00 | 380.00 | 405.00 |
| Adj. R-squared | 0.24 | 0.26 | 0.26 | 0.24 |
| Joint test: Trust only p-value | 0.50 | 0.40 | 0.47 | 0.45 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.00 | 0.00 | 0.00 | 0.00 |
| Joint test: Race p-value | 0.01 | 0.00 | 0.00 | 0.01 |
| Joint test: Region p-value | 0.09 | 0.08 | 0.06 | 0.09 |

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The cross-section exclusion checks are mixed. In most columns, the depression coefficient is not statistically different from zero; however, the 2018 lag is significant in the one-lag specification. The joint test for trust and trust-squared is not rejected across columns, so this cross-section setup is not the strongest environment for the exclusion argument.

Table 5.1.2 runs the same exclusion check using average returns over 2002–2022.

**Table 21:** Exclusion restriction: r5 average on controls + depression + Trust + Trust$^2$

| | No lag | One lag | Two lags | Ever depressed |
|---|---|---|---|---|
| Female | −0.04 | −0.03 | −0.03 | −0.04 |
| | (0.02) | (0.03) | (0.03) | (0.02) |
| Years of education | 0.00 | 0.00 | 0.01 | 0.00 |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| Married | −0.01 | −0.02 | −0.03 | −0.01 |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| NH Black | 0.02 | 0.02 | 0.02 | 0.02 |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| Hispanic | −0.09* | −0.08 | −0.08 | −0.09* |
| | (0.05) | (0.05) | (0.05) | (0.05) |
| NH Other | −0.02 | −0.05 | −0.05 | −0.02 |
| | (0.08) | (0.08) | (0.08) | (0.08) |
| In labor force | 0.10*** | 0.10*** | 0.09*** | 0.10*** |
| | (0.03) | (0.04) | (0.04) | (0.03) |
| Midwest | −0.01 | 0.00 | 0.00 | −0.01 |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| South | 0.03 | 0.06 | 0.05 | 0.03 |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| West | 0.13*** | 0.15*** | 0.15*** | 0.14*** |
| | (0.04) | (0.05) | (0.05) | (0.04) |
| Depression (no lag) | −0.00 | | | |
| | (0.01) | | | |
| Depression (lag 1) | | −0.00 | 0.00 | |
| | | (0.01) | (0.01) | |
| Depression (lag 2) | | | −0.00 | |
| | | | (0.01) | |
| Ever depressed | | | | −0.05 |
| | | | | (0.03) |
| Trust | 0.04*** | 0.04** | 0.04** | 0.04*** |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| Trust$^2$ | −0.00** | −0.00** | −0.00* | −0.00** |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| Constant | −0.02 | −0.04 | −0.04 | 0.03 |
| | (0.10) | (0.11) | (0.11) | (0.11) |
| Observations | 592.00 | 554.00 | 547.00 | 592.00 |
| Adj. R-squared | 0.09 | 0.08 | 0.08 | 0.09 |
| Joint test: Trust only p-value | 0.02 | 0.03 | 0.04 | 0.02 |
| Joint test: Age bins p-value | 0.12 | 0.22 | 0.28 | 0.13 |
| Joint test: Wealth deciles p-value | 0.10 | 0.12 | 0.15 | 0.08 |
| Joint test: Race p-value | 0.17 | 0.25 | 0.24 | 0.14 |
| Joint test: Region p-value | 0.00 | 0.00 | 0.00 | 0.00 |

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For average returns, the exclusion evidence is stronger: all depression variants are insignificant, while the trust coefficients preserve a hump-shaped pattern and the joint test on trust and trust-squared is rejected in all columns. This pattern is more consistent with depression serving as a predictor of trust rather than a direct predictor of long-run returns.

Table 5.1.2 repeats the exclusion check for the baseline panel specification (spec 1): returns to net wealth on controls, trust, trust-squared, and depression variants, with standard errors clustered by household. The panel uses the same control set and depression variants (no lag, one lag, two lags, ever depressed) as the cross-section and average specifications.

**Table 22:** Exclusion restriction: r5 panel (spec 1) on controls + depression + Trust + Trust$^2$

| | No lag | One lag | Two lags | Ever depressed |
|---|---|---|---|---|
| Female | $-0.03^{*}$ | $-0.03^{*}$ | $-0.03^{*}$ | $-0.03^{*}$ |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| Years of education | $-0.01^{***}$ | $-0.01^{***}$ | $-0.01^{***}$ | $-0.01^{***}$ |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| NH Black | $0.05^{*}$ | 0.04 | 0.04 | $0.05^{*}$ |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| Hispanic | $-0.01$ | $-0.02$ | $-0.03$ | $-0.01$ |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| NH Other | 0.01 | 0.00 | 0.02 | 0.01 |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| In labor force | $0.03^{*}$ | $0.03^{**}$ | $0.03^{*}$ | $0.03^{*}$ |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| Married | $-0.06^{***}$ | $-0.07^{***}$ | $-0.07^{***}$ | $-0.06^{***}$ |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| Born in U.S. | 0.03 | 0.03 | $0.04^{*}$ | 0.03 |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| Midwest | 0.00 | 0.01 | 0.01 | 0.01 |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| South | 0.04 | 0.03 | 0.03 | $0.04^{*}$ |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| West | $0.05^{*}$ | $0.05^{*}$ | $0.05^{*}$ | $0.05^{*}$ |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| Depression (no lag) | $-0.00$ | | | |
| | (0.00) | | | |
| Depression (lag 1) | | 0.00 | 0.01 | |
| | | (0.00) | (0.01) | |
| Depression (lag 2) | | | $-0.01$ | |
| | | | (0.01) | |
| Ever depressed | | | | $-0.01$ |
| | | | | (0.02) |
| Trust | $0.03^{***}$ | $0.03^{***}$ | $0.02^{**}$ | $0.03^{***}$ |
| | (0.01) | (0.01) | (0.01) | (0.01) |
| Trust$^2$ | $-0.00^{**}$ | $-0.00^{***}$ | $-0.00^{**}$ | $-0.00^{**}$ |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| Constant | $-0.53^{***}$ | $-0.53^{***}$ | $-0.42^{***}$ | $-0.53^{***}$ |
| | (0.06) | (0.06) | (0.06) | (0.06) |
| Observations | 2875.00 | 2754.00 | 2567.00 | 2899.00 |
| Adj. R-squared | 0.04 | 0.04 | 0.04 | 0.04 |
| Joint test: Trust only p-value | 0.00 | 0.00 | 0.03 | 0.00 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.02 | 0.02 | 0.04 | 0.01 |
| Joint test: Race p-value | 0.20 | 0.30 | 0.33 | 0.20 |
| Joint test: Region p-value | 0.10 | 0.21 | 0.28 | 0.10 |
| Joint test: Year p-value | 0.00 | 0.00 | 0.00 | 0.00 |

The panel exclusion checks are consistent with the average-returns results: depression variants are generally insignificant, and the joint test on trust and trust-squared is rejected, supporting the exclusion restriction in the panel setting.

#### 5.1.3 Instrument exogeneity (independence)

### 5.2 Second stage: predict returns with variation not explained by trust

Because trust enters the structural equation as both linear and quadratic terms, both *Trust* and $Trust^2$ are treated as endogenous. Let $Z_i$ denote the depression-based instrument variant ($Depression_{2020}$, $Depression_{2018}$, ($Depression_{2018}, Depression_{2016}$), or *EverDepressed*). Let $X$ denote the exogenous controls.

Here $\tilde{Z}_i$ is the instrument set used for the quadratic endogenous regressor (for example, transformations such as $Z_i^2$, and if needed interactions with exogenous controls).

**Cross section (2022 returns):**

$$Trust_{i,2020} = \pi_0 + X'_{i,2020}\pi_1 + Z'_i\pi_2 + v_i, \qquad Trust^2_{i,2020} = \rho_0 + X'_{i,2020}\rho_1 + \tilde{Z}'_i\rho_2 + u_i$$

$$r^{net}_{i,2022} = \alpha + \beta_1 \widehat{Trust}_{i,2020} + \beta_2 \widehat{Trust^2}_{i,2020} + X'_{i,2020}\gamma + \varepsilon_{i,2022}$$

**Average returns (2002–2022):**

$$Trust_{i,2020} = \pi_0 + X'_{i,2020}\pi_1 + Z'_i\pi_2 + v_i, \qquad Trust^2_{i,2020} = \rho_0 + X'_{i,2020}\rho_1 + \tilde{Z}'_i\rho_2 + u_i$$

$$\bar{r}^{net}_{i,2002:2022} = \alpha + \beta_1 \widehat{Trust}_{i,2020} + \beta_2 \widehat{Trust^2}_{i,2020} + X'_{i,2020}\gamma + \bar{\varepsilon}_i$$

**Panel (baseline pooled panel specification):**

$$Trust_{i,2020} = \pi_0 + X'_{i,2020}\pi_1 + Z'_i\pi_2 + v_i, \qquad Trust^2_{i,2020} = \rho_0 + X'_{i,2020}\rho_1 + \tilde{Z}'_i\rho_2 + u_i$$

$$r^{net}_{i,t} = \alpha + \beta_1 \widehat{Trust}_{i,2020} + \beta_2 \widehat{Trust^2}_{i,2020} + X'_{i,t}\gamma + \varepsilon_{i,t}, \quad t = 2002, \ldots, 2022$$

## 6 Extensions

In this section, I describe the data and accompanying statistical model used to understand the role of *financial literacy* in the current story of an empirical relationship between trust and returns.

Following the argument that trust is at the heart of financial agreements, the extent to which investors trust the financial institutions they make contracts with should be the sense in which trust effects returns. To test this, I redo the analysis from the main text using an aggregate measure of trust in financial institutions.

### 6.1 Financial literacy and returns

Financial literacy can be an important factor when trying to understand the persistent component of returns. The literature on measuring financial literacy using survey data has agreed on three questions (the "Big Three") to measure financial literacy for respondents, and the HRS 2020 wave adopts the folllowing variation of these questions as a part of its survey:

1. **Interest**
   Suppose you had $100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?
   *(a) Less than $102 (b) Exactly $102 (c) More than $102*

2. **Inflation**
   Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?
   *(a) More than today (b) Exactly the same as today (c) Less than today*

3. **Risk Diversification**
   Please tell us whether this statement is true or false: "Buying a single company's stock usually provides a safer return than a stock mutual fund."
   *(a) True (b) False*

[12]

### 6.1.1 Descriptive statistics

Again, the persistent component of returns suggests that there is substantial variation in returns across individuals on average. Uncovering what drives these differences can be of particular interest to other literature aimed at modeling household behavior and decision problems.

To begin in this direction, consider in the summary for the proportion of the sample that answered each of the questions correctly in table 6.1.1.

**Table 23:** .

| Variable | N | Mean |
|---|---|---|
| Interest | 1302 | 0.74 |
| Inflation | 1273 | 0.76 |
| Risk | 1072 | 0.75 |

The "Literacy score" refers to a score between 0 and 3 showing how many of the three questions each respondent got correctly. As see from the table 6.1.1, almost half of the individuals answered all three correctly.

[12] "DK" (Don't Know) and "RF" (Refused) are possible non-responses for each of the questions.

**Table 24:** Financial literacy: Literacy score (0–3) frequency

| Literacy score | N | Pct |
|---|---|---|
| 0 | 25 | 2.4 |
| 1 | 150 | 14.4 |
| 2 | 364 | 35.0 |
| 3 | 500 | 48.1 |

Table 6.1.1 is a breakdown of the sample performance on the financial literacy score by groupings relevant to the control variables of the empirical analysis (gender, education, age, and race).

**Table 25:** Financial literacy score by demographics

**Panel A: By gender**

| Group | N | Mean |
|---|---|---|
| Male | 471 | 2.33 |
| Female | 568 | 2.25 |

**Panel B: By age**

| Age | N | Mean |
|---|---|---|
| 25–49 | 22 | 1.56 |
| 50–54 | 41 | 2.49 |
| 55–59 | 207 | 2.20 |
| 60–64 | 217 | 2.34 |
| 65–69 | 212 | 2.33 |
| 70–74 | 130 | 2.39 |
| 75–79 | 92 | 2.38 |
| 80–84 | 80 | 2.08 |
| 85–89 | 37 | 2.30 |

**Panel C: By race**

| Group | N | Mean |
|---|---|---|
| White (NH) | 619 | 2.45 |
| Black (NH) | 209 | 2.03 |
| Hispanic | 138 | 2.01 |
| Other (NH) | 72 | 2.19 |

**Panel D: By education**

| Education | N | Mean |
|---|---|---|
| No hs | 147 | 2.01 |
| Hs | 267 | 2.01 |
| Some college | 269 | 2.35 |
| 4yr degree | 177 | 2.54 |
| Grad | 173 | 2.60 |

Along with the aggregated literacy score, the inflation question has the best performance among the measures in terms of the statistical analysis. I take a look at the same breakdown but for the "Inflation" in table 6.1.1.

**Table 26:** Financial literacy (inflation question) by demographics

**Panel A: By gender**

| Group | N | Mean |
|---|---|---|
| Male | 531 | 0.79 |
| Female | 742 | 0.75 |

**Panel B: By age**

| Age | N | Mean |
|---|---|---|
| 25–49 | 24 | 0.40 |
| 50–54 | 50 | 0.74 |
| 55–59 | 249 | 0.74 |
| 60–64 | 256 | 0.79 |
| 65–69 | 250 | 0.78 |
| 70–74 | 154 | 0.81 |
| 75–79 | 123 | 0.85 |
| 80–84 | 120 | 0.68 |
| 85–89 | 46 | 0.76 |

**Panel C: By race**

| Group | N | Mean |
|---|---|---|
| White (NH) | 729 | 0.82 |
| Black (NH) | 270 | 0.67 |
| Hispanic | 192 | 0.67 |
| Other (NH) | 81 | 0.77 |

**Panel D: By education**

| Education | N | Mean |
|---|---|---|
| No hs | 211 | 0.67 |
| Hs | 330 | 0.74 |
| Some college | 340 | 0.75 |
| 4yr degree | 200 | 0.83 |
| Grad | 186 | 0.87 |

Table 6.1.1 shows the cross correlations between the financial literacy measures and the general measure of trust. There is little correlation between trust and any of the financial literacy variables. More worrisome is that there is little correlation amongst the financial literacy variables. If they are all measuring the same thing, there should be some correlation.

**Table 27:** .

| | Interest | Inflation | Risk div. | Literacy score | General trust |
|---|---|---|---|---|---|
| Interest | 1.00 | 0.03 | 0.03 | 0.60 | -0.00 |
| Inflation | 0.03 | 1.00 | 0.08 | 0.59 | -0.03 |
| Risk div. | 0.03 | 0.08 | 1.00 | 0.62 | 0.04 |
| Literacy score | 0.60 | 0.59 | 0.62 | 1.00 | -0.00 |
| General trust | -0.00 | -0.03 | 0.04 | -0.00 | 1.00 |

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### 6.1.2 Results

Not only is the financial literacy measure a potential determinant of differences in average returns across individuals, but it also may help to explain the variation in returns to net wealth. To consider these effects, I run each of the models of interest (cross section, average, panel specifications 1,2, and 3) with either the three financial literacy questions (interest, inflation, risk diversification) or the aggregated financial literacy score. The analysis concludes that the inflation question and the financial literacy score have predictive power for returns.

At the cross section, both "Inflation" and "Literacy score" are positively correlated with returns to net wealth. This can be seen in table 6.1.2. Notably, both coefficients reject the null hypothesis whether or not trust is included in the regression. The coefficients for gender, education, and race retain their predictive power for returns in this setting. The Wald tests are rejected for the coefficients on age bins, wealth deciles, and race categories.

**Table 28:** Cross section: r5 returns (5 percent wins) on financial literacy and trust

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Female | $-0.11^{**}$ | $-0.12^{**}$ | $-0.11^{**}$ | $-0.12^{**}$ |
| | (0.05) | (0.05) | (0.05) | (0.05) |
| Years education | $0.02^{**}$ | $0.02^{**}$ | $0.02^{**}$ | $0.02^{*}$ |
| | (0.01) | (0.01) | (0.01) | (0.01) |
| In labor force | $0.12^{**}$ | $0.11^{**}$ | $0.12^{**}$ | $0.11^{**}$ |
| | (0.05) | (0.05) | (0.05) | (0.05) |
| Married | 0.06 | 0.06 | 0.06 | 0.06 |
| | (0.05) | (0.05) | (0.06) | (0.06) |
| Born in U.S. | 0.03 | 0.03 | 0.02 | 0.03 |
| | (0.06) | (0.06) | (0.06) | (0.06) |
| NH Black | $-0.19^{**}$ | $-0.19^{**}$ | $-0.18^{**}$ | $-0.18^{**}$ |
| | (0.08) | (0.08) | (0.08) | (0.08) |
| Hispanic | $-0.14$ | $-0.14$ | $-0.13$ | $-0.14$ |
| | (0.09) | (0.09) | (0.09) | (0.09) |
| NH Other | 0.06 | 0.07 | 0.07 | 0.07 |
| | (0.11) | (0.11) | (0.11) | (0.11) |
| Interest | 0.07 | | 0.07 | |
| | (0.07) | | (0.07) | |
| Inflation | $0.15^{**}$ | | $0.16^{**}$ | |
| | (0.06) | | (0.06) | |
| Risk div | 0.01 | | 0.01 | |
| | (0.06) | | (0.06) | |
| Literacy score | | $0.07^{*}$ | | $0.08^{**}$ |
| | | (0.04) | | (0.04) |
| General trust | | | 0.01 | 0.01 |
| | | | (0.03) | (0.04) |
| (General trust)$^2$ | | | 0.00 | $-0.00$ |
| | | | (0.00) | (0.00) |
| _cons | $-0.59^{***}$ | $-0.57^{***}$ | $-0.61^{***}$ | $-0.58^{***}$ |
| | (0.23) | (0.22) | (0.23) | (0.22) |
| Observations | 319.00 | 319.00 | 319.00 | 319.00 |
| Adj. R-squared | 0.25 | 0.25 | 0.25 | 0.25 |
| Joint test: Trust p-value | . | . | 0.69 | 0.80 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.00 | 0.00 | 0.00 | 0.00 |
| Joint test: Race p-value | 0.06 | 0.05 | 0.08 | 0.06 |

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For average returns over the period, only the inflation question is explanatory for returns and retains its predictive power when trust is included. However, unlike at the cross section, the Wald test on the linear and quadratic trust terms is rejected here for both "Inflation" and "Literacy score". Interestingly, I fail to reject the Wald tests for the coefficients on age bins, wealth deciles, and race

categories in this setting based on the large p-values found in the table 6.1.2.

**Table 29:** Average r5 returns (5 percent wins) on financial literacy and trust

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Female | −0.04 | −0.04 | −0.05 | −0.05* |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| Years education | 0.00 | 0.00 | 0.00 | 0.00 |
| | (0.01) | (0.01) | (0.01) | (0.01) |
| In labor force | 0.08** | 0.08** | 0.08** | 0.07** |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| Married | 0.02 | 0.02 | 0.02 | 0.02 |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| Born in U.S. | −0.03 | −0.03 | −0.04 | −0.04 |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| NH Black | −0.03 | −0.03 | −0.01 | −0.01 |
| | (0.05) | (0.05) | (0.05) | (0.05) |
| Hispanic | −0.09 | −0.10* | −0.08 | −0.09 |
| | (0.06) | (0.06) | (0.06) | (0.06) |
| NH Other | 0.02 | 0.02 | 0.03 | 0.03 |
| | (0.07) | (0.07) | (0.07) | (0.07) |
| Interest | −0.00 | | 0.00 | |
| | (0.03) | | (0.03) | |
| Inflation | 0.08** | | 0.09** | |
| | (0.04) | | (0.04) | |
| Risk div | −0.03 | | −0.03 | |
| | (0.04) | | (0.04) | |
| Literacy score | | 0.01 | | 0.02 |
| | | (0.02) | | (0.02) |
| General trust | | | 0.03 | 0.03 |
| | | | (0.02) | (0.02) |
| (General trust)$^2$ | | | −0.00 | −0.00 |
| | | | (0.00) | (0.00) |
| _cons | 0.11 | 0.11 | 0.09 | 0.09 |
| | (0.12) | (0.12) | (0.12) | (0.12) |
| Observations | 440.00 | 440.00 | 440.00 | 440.00 |
| Adj. R-squared | 0.04 | 0.03 | 0.05 | 0.04 |
| Joint test: Trust p-value | . | . | 0.04 | 0.07 |
| Joint test: Age bins p-value | 0.23 | 0.19 | 0.42 | 0.49 |
| Joint test: Wealth deciles p-value | 0.33 | 0.25 | 0.24 | 0.17 |
| Joint test: Race p-value | 0.39 | 0.31 | 0.46 | 0.37 |

.

The results for the third panel regression specification 6.1.2 are the most interesting. In the second stage of this final panel specification *without* financial literacy, trust lacks significance re-

garding the explanatory power of variation in trust for variation in the estimated fixed effects. This was reflected in the large p-values for the tests on the individual coefficients, as well as for the joint Wald test.

**Table 30:** Panel Spec 3 (FE, 2nd stage): r5 returns on financial literacy and trust

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Years education | 0.03*** | 0.03*** | 0.03*** | 0.03*** |
| | (0.00) | (0.00) | (0.00) | (0.00) |
| Female | −0.07*** | −0.07*** | −0.07*** | −0.07*** |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| NH Black | −0.19*** | −0.19*** | −0.19*** | −0.19*** |
| | (0.03) | (0.03) | (0.03) | (0.03) |
| Hispanic | −0.14*** | −0.14*** | −0.14*** | −0.14*** |
| | (0.05) | (0.05) | (0.05) | (0.05) |
| NH Other | −0.06 | −0.06 | −0.05 | −0.06 |
| | (0.07) | (0.07) | (0.07) | (0.07) |
| Born in U.S. | −0.09** | −0.09** | −0.09** | −0.09** |
| | (0.04) | (0.04) | (0.04) | (0.04) |
| Interest | 0.07** | | 0.07** | |
| | (0.03) | | (0.03) | |
| Inflation | 0.05 | | 0.04 | |
| | (0.04) | | (0.03) | |
| Risk div | 0.07** | | 0.07** | |
| | (0.03) | | (0.03) | |
| Literacy score | | 0.06*** | | 0.06*** |
| | | (0.02) | | (0.02) |
| General trust | | | 0.04** | 0.03** |
| | | | (0.02) | (0.02) |
| (General trust)$^2$ | | | −0.00** | −0.00** |
| | | | (0.00) | (0.00) |
| _cons | −0.31*** | −0.31*** | −0.36*** | −0.36*** |
| | (0.09) | (0.09) | (0.09) | (0.09) |
| Observations | 417.00 | 417.00 | 417.00 | 417.00 |
| Adj. R-squared | 0.27 | 0.27 | 0.27 | 0.27 |
| Joint test: Trust p-value | . | . | 0.07 | 0.10 |
| Joint test: Race p-value | 0.00 | 0.00 | 0.00 | 0.00 |

.

When the financial literacy variables are included in the second stage ("Big Three" or financial literacy score), not only are those coefficients signficant, but the individual trust coefficients are now significant as well. However, the story is not all good – the signs on the estimated coefficients for the linear trust term are negative (so the maximal economic performance does not make sense in this setting).

## 6.2 Trust in financial institutions

To focus specifically on the domain of trust most directly related to financial transactions, similar to Haran Rosen, Lusardi, and Mitchell 2025 I construct composite measures of trust in financial institutions, aggregating responses related to trust in banks, mutual funds, and financial advisors. This targeted measure of trust is more theoretically aligned with the mechanisms through which interpersonal trust may influence realized investment returns. I present results using two aggregation strategies: (i) the simple average of the three trust scores, and (ii) the PCA analysis including those three variables and the general trust measure.

### 6.2.1 Composite trust score

Table 6.2.1 shows average trust levels for each of the indidivual trust components used to create the "Trust in Fin. Institutions" variable. The sample seems more trusting of banks than financial advisors or mutual funds, which pushes up the average trust level among the three.

**Table 31:** Financial-institutional trust: summary (2020)

| Variable | N | Mean |
|---|---|---|
| Banks | 990 | 5.35 |
| Financial advisors | 1080 | 4.29 |
| Mutual funds | 1072 | 4.29 |
| Trust in Fin. Institutions | 1234 | 4.78 |

Table 6.2.1 reports pairwise correlations among general trust and the three financial-institution trust components (banks, financial advisors, mutual funds). This helps benchmark how closely the general trust measure co-moves with the institution-specific trust variables used in the extension.

**Table 32:** .

| | General trust | Banks | Financial advisors | Mutual funds |
|---|---|---|---|---|
| General trust | 1.000 | 0.363 | 0.333 | 0.332 |
| Banks | 0.363 | 1.000 | 0.546 | 0.411 |
| Financial advisors | 0.333 | 0.546 | 1.000 | 0.636 |
| Mutual funds | 0.332 | 0.411 | 0.636 | 1.000 |

The results for this section can be summarized as follows. For the 2022 cross-section and for the average over the period, trust in financial institutions lacks explanatory power for returns to net wealth. This is reflected by the tests on the individual coefficients as well as the joint Wald test.

The panel regression results are slightly more favorable. For the baseline regression, table 6.2.1 shows that the joint Wald test for the linear and quadratic trust term is rejected. The joint tests for age bins, wealth deciles, and race categories behaves similarly.

**Table 33:** Panel Spec 1 (pooled): r5 returns on financial-institutional trust

| | (1) | (2) | (3) |
|---|---|---|---|
| Years education | −0.00 | −0.00 | −0.00 |
| | (0.00) | (0.00) | (0.00) |
| Female | −0.01 | −0.01 | −0.01 |
| | (0.01) | (0.01) | (0.01) |
| NH Black | 0.07*** | 0.07*** | 0.07*** |
| | (0.02) | (0.02) | (0.02) |
| Hispanic | 0.06** | 0.05** | 0.05** |
| | (0.03) | (0.03) | (0.03) |
| NH Other | 0.03 | 0.03 | 0.03 |
| | (0.03) | (0.03) | (0.03) |
| In labor force | 0.03** | 0.03** | 0.03** |
| | (0.01) | (0.01) | (0.01) |
| Married | −0.06*** | −0.06*** | −0.06*** |
| | (0.01) | (0.01) | (0.01) |
| Born in U.S. | 0.01 | 0.01 | 0.01 |
| | (0.02) | (0.02) | (0.02) |
| Trust | | −0.01** | 0.00 |
| | | (0.00) | (0.01) |
| $\text{Trust}^2$ | | | −0.00 |
| | | | (0.00) |
| _cons | −0.55*** | −0.54*** | −0.55*** |
| | (0.04) | (0.04) | (0.04) |
| Observations | 4219.00 | 4219.00 | 4219.00 |
| Adj. R-squared | 0.02 | 0.02 | 0.02 |
| Joint test: Trust p-value | . | . | 0.06 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Race p-value | 0.00 | 0.01 | 0.01 |
| Joint test: Region p-value | 0.07 | 0.10 | 0.10 |

.

I turn attention to the empirical relationship between trust in financial institutions and returns when controlling for risk exposure in table 6.2.1. The joint Wald test on the trust coefficients is rejected here as well; however, the coefficient on the linear trust term is negative. This suggests that the critical value for trust in financial institutions here does not maximize returns to net wealth.

**Table 34:** Panel Spec 2 (share x year): r5 returns on financial-institutional trust

| | (1) | (2) | (3) |
|---|---|---|---|
| Years education | $-0.01^{**}$ | $-0.01^{**}$ | $-0.01^{**}$ |
| | (0.00) | (0.00) | (0.00) |
| Female | $-0.01$ | $-0.01$ | $-0.01$ |
| | (0.01) | (0.01) | (0.01) |
| NH Black | $0.04^{**}$ | $0.04^{*}$ | $0.04^{*}$ |
| | (0.02) | (0.02) | (0.02) |
| Hispanic | 0.02 | 0.02 | 0.02 |
| | (0.03) | (0.03) | (0.03) |
| NH Other | 0.03 | 0.03 | 0.03 |
| | (0.03) | (0.03) | (0.03) |
| In labor force | $0.03^{**}$ | $0.03^{**}$ | $0.03^{**}$ |
| | (0.01) | (0.01) | (0.01) |
| Married | $-0.05^{***}$ | $-0.05^{***}$ | $-0.05^{***}$ |
| | (0.01) | (0.01) | (0.01) |
| Born in U.S. | 0.01 | 0.01 | 0.01 |
| | (0.02) | (0.02) | (0.02) |
| Trust | | $-0.01^{**}$ | $-0.00$ |
| | | (0.00) | (0.01) |
| $\text{Trust}^2$ | | | $-0.00$ |
| | | | (0.00) |
| _cons | $-0.56^{***}$ | $-0.55^{***}$ | $-0.56^{***}$ |
| | (0.20) | (0.20) | (0.20) |
| Observations | 4219.00 | 4219.00 | 4219.00 |
| Adj. R-squared | 0.07 | 0.07 | 0.07 |
| Joint test: Trust p-value | . | . | 0.06 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Race p-value | 0.19 | 0.25 | 0.26 |
| Joint test: Region p-value | 0.03 | 0.04 | 0.04 |
| Joint test: Year p-value | 0.00 | 0.00 | 0.00 |

.

Lastly, results for the final panel specification on trust in financial institutions allowing for individual fixed effects can be seen in table 6.2.1. Interestingly, trust in financial institutions has strong predictive power for the individual fixed effect. In fact, the relationship appears to be hump-shaped: those with higher estimated fixed effects have moderate trust levels.

**Table 35:** Panel Spec 3 (FE, 2nd stage): r5 returns on financial-institutional trust

| | (1) | (2) | (3) |
|---|---|---|---|
| Years education | 0.03*** | 0.03*** | 0.03*** |
| | (0.00) | (0.00) | (0.00) |
| Female | −0.05** | −0.05*** | −0.05** |
| | (0.02) | (0.02) | (0.02) |
| NH Black | −0.22*** | −0.20*** | −0.19*** |
| | (0.03) | (0.03) | (0.03) |
| Hispanic | −0.17*** | −0.15*** | −0.16*** |
| | (0.03) | (0.03) | (0.03) |
| NH Other | −0.06 | −0.04 | −0.04 |
| | (0.06) | (0.06) | (0.06) |
| Born in U.S. | −0.06 | −0.06 | −0.06 |
| | (0.04) | (0.04) | (0.04) |
| Trust | | 0.02*** | 0.06*** |
| | | (0.00) | (0.02) |
| $\text{Trust}^2$ | | | −0.00*** |
| | | | (0.00) |
| _cons | −0.21*** | −0.31*** | −0.36*** |
| | (0.06) | (0.06) | (0.07) |
| Observations | 795.00 | 795.00 | 795.00 |
| Adj. R-squared | 0.23 | 0.26 | 0.26 |
| Joint test: Trust p-value | . | . | 0.00 |
| Joint test: Race p-value | 0.00 | 0.00 | 0.00 |

.

To assess whether the hump-shaped pattern implied by table 6.2.1 is visible in the data, figure 35 plots the estimated person fixed effect against trust in financial institutions. The scatter does not show a clear inverted-U shape, suggesting caution in interpreting the quadratic fit as strong visual evidence of a hump-shaped relationship.

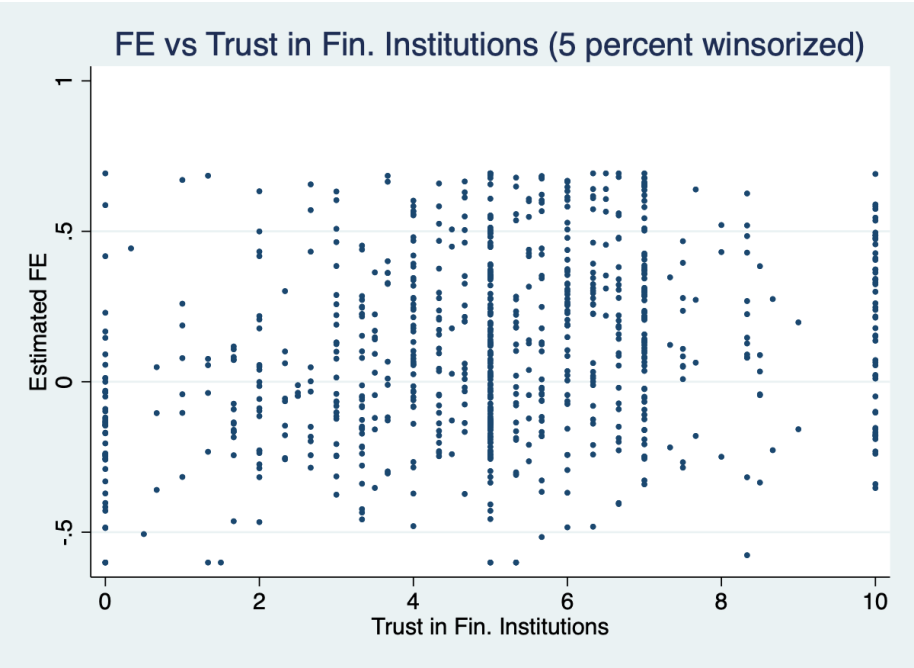


**Figure 35:** Estimated fixed effects and trust in financial institutions

### 6.2.2 Principal component analysis

Principal component analysis (PCA) is a dimensionality reduction technique that transforms a set of correlated variables into a smaller number of uncorrelated components, each capturing a portion of the original variance. In the context of survey-based econometric analysis, PCA helps synthesize information from related variables—in this case, trust in financial advisors, mutual funds, and banks—into a single latent index. Including general trust alongside these financial trust variables in the PCA allows us to examine whether broader interpersonal trust co-moves with institutional trust and to what extent it contributes to the underlying factor structure.

Figure 36 displays the scree plot from the PCA analysis. The first component accounts for a significant share of the total variance, with a sharp drop in explained variance after the first eigenvalue. This suggests a one-factor structure may be appropriate, capturing a dominant dimension of trust in financial institutions. The relatively flat tail beyond the first component supports the decision to focus on the first principal component for constructing a composite measure.

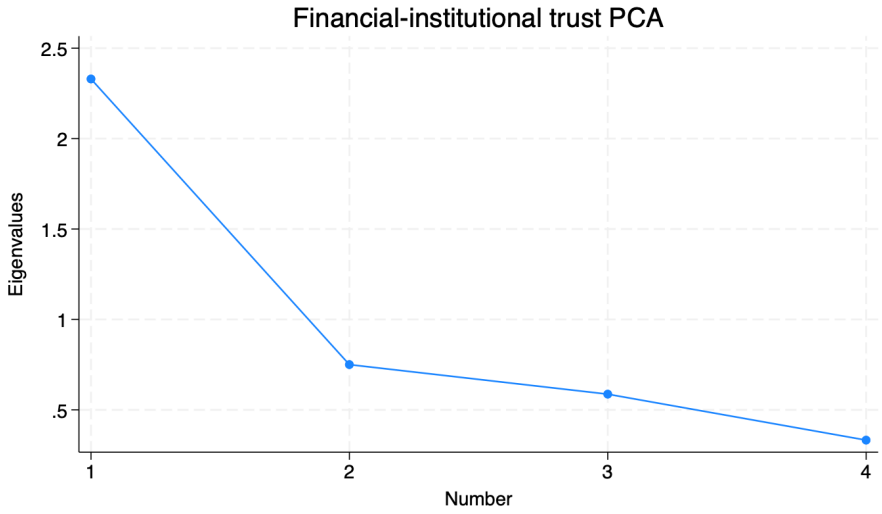


**Figure 36:** Financial-institutional trust PCA: scree plot

Table 6.2.2 presents the loadings for each trust variable on the first two components. Loadings represent the contribution of each original variable to the new principal components. In Component 1, all four trust variables load positively and similarly, indicating they move together and reflect a common latent factor—likely overall financial trust. Component 2, in contrast, differentiates general trust from the institutional trust variables, with a notably higher loading for general trust. This implies that while general trust is correlated, it may capture a broader dimension of trust beyond financial institutions.

**Table 36:** Financial-institutional trust PCA: loadings by variable (4 vars: general, banks, advisors, mutual funds)

| Variable | PC1 | PC2 |
|---|---|---|
| General trust | 0.407 | 0.879 |
| Banks | 0.501 | 0.012 |
| Financial advisors | 0.557 | -0.334 |
| Mutual funds | 0.523 | -0.340 |

For the first component, the average-return results in table 6.2.2 are only modestly supportive. In the quadratic specification, the trust-squared term is negative and statistically significant, while the linear trust term is small and not statistically different from zero. The joint Wald test on the trust terms is rejected which suggests some nonlinearity, but not a strong hump-shaped pattern.

**Table 37:** Average r5 returns (5 percent wins) on financial-institutional trust (PC1)

| | (1) | (2) | (3) |
|---|---|---|---|
| Female | −0.05 | −0.05 | −0.05 |
| | (0.03) | (0.03) | (0.03) |
| Years education | 0.01 | 0.01 | 0.01 |
| | (0.01) | (0.01) | (0.01) |
| In labor force | 0.11*** | 0.11*** | 0.10** |
| | (0.04) | (0.04) | (0.04) |
| Married | −0.02 | −0.02 | −0.03 |
| | (0.04) | (0.04) | (0.04) |
| Born in U.S. | 0.01 | 0.01 | 0.01 |
| | (0.05) | (0.05) | (0.05) |
| NH Black | 0.00 | 0.00 | 0.01 |
| | (0.05) | (0.05) | (0.05) |
| Hispanic | −0.02 | −0.02 | −0.02 |
| | (0.06) | (0.06) | (0.06) |
| NH Other | 0.02 | 0.02 | 0.02 |
| | (0.09) | (0.09) | (0.09) |
| Trust | | −0.00 | 0.00 |
| | | (0.01) | (0.01) |
| $\text{Trust}^2$ | | | −0.01** |
| | | | (0.01) |
| _cons | 0.04 | 0.04 | 0.12 |
| | (0.12) | (0.12) | (0.12) |
| Observations | 411.00 | 411.00 | 411.00 |
| Adj. R-squared | 0.05 | 0.05 | 0.06 |
| Joint test: Trust p-value | . | . | 0.09 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.02 | 0.02 | 0.01 |
| Joint test: Race p-value | 0.99 | 0.99 | 0.97 |

.

The FE second-stage results for the first component in table 6.2.2 are stronger. The linear trust coefficient is positive and significant, the quadratic term is negative and significant, and the joint Wald test is strongly rejected. This is consistent with a hump-shaped relationship, where moderate levels of financial-institutional trust are associated with higher estimated fixed effects.

**Table 38:** Panel Spec 3 (FE, 2nd stage): r5 returns on financial-institutional trust (PC1)

| | (1) | (2) | (3) |
|---|---|---|---|
| Years education | 0.02*** | 0.02*** | 0.02*** |
| | (0.00) | (0.00) | (0.00) |
| Female | −0.04 | −0.06** | −0.06* |
| | (0.03) | (0.03) | (0.03) |
| NH Black | −0.24*** | −0.22*** | −0.21*** |
| | (0.04) | (0.04) | (0.04) |
| Hispanic | −0.19*** | −0.18*** | −0.19*** |
| | (0.05) | (0.05) | (0.05) |
| NH Other | −0.07 | −0.05 | −0.05 |
| | (0.07) | (0.07) | (0.08) |
| Born in U.S. | −0.07 | −0.07 | −0.08 |
| | (0.06) | (0.06) | (0.06) |
| Trust | | 0.04*** | 0.05*** |
| | | (0.01) | (0.01) |
| $\text{Trust}^2$ | | | −0.01** |
| | | | (0.00) |
| _cons | −0.03 | −0.01 | 0.03 |
| | (0.07) | (0.07) | (0.07) |
| Observations | 384.00 | 384.00 | 384.00 |
| Adj. R-squared | 0.17 | 0.20 | 0.21 |
| Joint test: Trust p-value | . | . | 0.00 |
| Joint test: Race p-value | 0.00 | 0.00 | 0.00 |

.

The regression results when using the second component differ from the first component. In table 6.2.2, the linear trust term is positive and statistically significant, while the quadratic term is not significant. Although the joint Wald test is rejected, the fitted pattern is closer to a positive linear association than to a hump shape, so there is no clear interior trust level that maximizes returns.

**Table 39:** Average r5 returns (5 percent wins) on financial-institutional trust (PC2)

| | (1) | (2) | (3) |
|---|---|---|---|
| Female | −0.05 | −0.05 | −0.05 |
| | (0.03) | (0.03) | (0.03) |
| Years education | 0.01 | 0.01 | 0.01 |
| | (0.01) | (0.01) | (0.01) |
| In labor force | 0.11*** | 0.11*** | 0.11*** |
| | (0.04) | (0.04) | (0.04) |
| Married | −0.02 | −0.02 | −0.02 |
| | (0.04) | (0.04) | (0.04) |
| Born in U.S. | 0.01 | −0.00 | −0.00 |
| | (0.05) | (0.05) | (0.05) |
| NH Black | 0.00 | 0.01 | 0.01 |
| | (0.05) | (0.05) | (0.05) |
| Hispanic | −0.02 | −0.01 | −0.01 |
| | (0.06) | (0.06) | (0.06) |
| NH Other | 0.02 | 0.02 | 0.02 |
| | (0.09) | (0.09) | (0.09) |
| Trust | | 0.05*** | 0.05*** |
| | | (0.02) | (0.02) |
| $\text{Trust}^2$ | | | 0.00 |
| | | | (0.01) |
| _cons | 0.04 | 0.10 | 0.09 |
| | (0.12) | (0.12) | (0.12) |
| Observations | 411.00 | 411.00 | 411.00 |
| Adj. R-squared | 0.05 | 0.07 | 0.07 |
| Joint test: Trust p-value | . | . | 0.02 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.02 | 0.03 | 0.03 |
| Joint test: Race p-value | 0.99 | 0.97 | 0.97 |

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A similar pattern appears in the pooled panel specification in table 6.2.2. The linear trust coefficient is positive and significant, the trust-squared coefficient is not significant, and the joint trust test is rejected. Again, this points to a mostly increasing relationship over the support of the data rather than a hump-shaped one.

**Table 40:** Panel Spec 1 (pooled): r5 returns on financial-institutional trust (PC2)

| | (1) | (2) | (3) |
|---|---|---|---|
| Years education | −0.00 | −0.00 | −0.00 |
| | (0.00) | (0.00) | (0.00) |
| Female | −0.02 | −0.02 | −0.02 |
| | (0.02) | (0.02) | (0.02) |
| NH Black | 0.04 | 0.05* | 0.05* |
| | (0.03) | (0.03) | (0.03) |
| Hispanic | 0.03 | 0.03 | 0.03 |
| | (0.04) | (0.04) | (0.04) |
| NH Other | −0.03 | −0.02 | −0.02 |
| | (0.04) | (0.04) | (0.04) |
| In labor force | 0.03 | 0.03 | 0.03 |
| | (0.02) | (0.02) | (0.02) |
| Married | −0.07*** | −0.07*** | −0.07*** |
| | (0.02) | (0.02) | (0.02) |
| Born in U.S. | 0.03 | 0.02 | 0.02 |
| | (0.03) | (0.03) | (0.03) |
| Trust | | 0.03*** | 0.03*** |
| | | (0.01) | (0.01) |
| $\text{Trust}^2$ | | | −0.00 |
| | | | (0.01) |
| _cons | −0.58*** | −0.54*** | −0.54*** |
| | (0.06) | (0.06) | (0.06) |
| Observations | 1864.00 | 1864.00 | 1864.00 |
| Adj. R-squared | 0.02 | 0.02 | 0.02 |
| Joint test: Trust p-value | . | . | 0.02 |
| Joint test: Age bins p-value | 0.00 | 0.00 | 0.00 |
| Joint test: Wealth deciles p-value | 0.08 | 0.06 | 0.06 |
| Joint test: Race p-value | 0.39 | 0.27 | 0.27 |
| Joint test: Region p-value | 0.04 | 0.03 | 0.03 |

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Finally, table 6.2.2 shows that in the FE second stage for the second component, both trust coefficients are negative and statistically significant, and the joint Wald test is strongly rejected (p-value 0.00). With a negative linear and negative quadratic term, this specification implies a decreasing relationship over the relevant range, so it does not imply an interior trust-maximizing point for returns.

**Table 41:** Panel Spec 3 (FE, 2nd stage): r5 returns on financial-institutional trust (PC2)

| | (1) | (2) | (3) |
|---|---|---|---|
| Years education | 0.02*** | 0.02*** | 0.02*** |
| | (0.00) | (0.00) | (0.00) |
| Female | −0.04 | −0.05 | −0.05 |
| | (0.03) | (0.03) | (0.03) |
| NH Black | −0.24*** | −0.26*** | −0.25*** |
| | (0.04) | (0.04) | (0.04) |
| Hispanic | −0.19*** | −0.19*** | −0.19*** |
| | (0.05) | (0.05) | (0.05) |
| NH Other | −0.07 | −0.08 | −0.07 |
| | (0.07) | (0.07) | (0.07) |
| Born in U.S. | −0.07 | −0.05 | −0.04 |
| | (0.06) | (0.06) | (0.06) |
| Trust | | −0.07*** | −0.06*** |
| | | (0.02) | (0.02) |
| $\text{Trust}^2$ | | | −0.02* |
| | | | (0.01) |
| _cons | −0.03 | −0.04 | −0.00 |
| | (0.07) | (0.07) | (0.07) |
| Observations | 384.00 | 384.00 | 384.00 |
| Adj. R-squared | 0.17 | 0.20 | 0.20 |
| Joint test: Trust p-value | . | . | 0.00 |
| Joint test: Race p-value | 0.00 | 0.00 | 0.00 |

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